BMI and documented hypercapnia in acute care

Expanded scientific report for investigator review

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4 Interpretation 54

Protected research draft. Investigator interpretation and scientific approval remain pending.

Saved run: curve-analyze-20261026T001917-75549. Analysis: bmi_hypercapnia_rfs_curve_audit_v9.

Source: identity_bound_source_declarations_passed; analysis completed: TRUE. Review states in the immutable bundle are recorded at estimation time; later signed review receipts are separate.

1 Introduction

We describe BMI–hypercapnia relationships within a selected acute-care cohort while exposing how historical source-entry pathways shape the observed population. These pathways record representation in reason-for-suspicion (RFS) source files; they do not necessarily identify every clinical criterion present at an encounter. Separating entry routes can change the observed association itself and does not eliminate outcome-dependent selection.

2 Methods

2.1 Source, eligibility and fixed index

The authenticated original FULL_DATA encounter list defines candidate visits. Adults had a 2022 emergency or inpatient encounter and any available valid BMI, without an obesity restriction. Age uses the inherited birth-year proxy. Each patient contributes the first qualifying encounter before gas availability and adjustment completeness. Earlier unresolved eligibility, conflicting evidence and earliest-date ties remain unresolved. Later encounters never replace an obesity-only index.

BMI follows the approved encounter/prior reported-value hierarchy, followed by calculation from observed height and weight, with inherited rounding and validity rules. One-unit diagnostic bins around BMI 40 and 50 describe measurement and selection; they are not a regression-discontinuity design. No statistical BMI imputation is performed. Available BMI is not necessarily contemporaneous with the outcome. The audit reports measurement methods, lags, unknown chronology and consequences of the existing rules. Signed lags equal encounter-start date minus measurement date: positive means prior, zero means the same calendar day, and negative means later during the encounter. Component gaps are absolute differences between height and weight dates.

2.1.1 Usable gas evidence and ascertainment

A usable result is at least one captured ordinary, unadjusted PCO_2 measurement linked to the selected encounter, with authenticated arterial or venous specimen assignment, reviewed units convertible to mm Hg, and strict validity limits of 5 < arterial PCO_2 <200 or 10 < venous PCO_2 <250 mm Hg. Normal and elevated results both establish usable evidence; neither hypercapnia nor pH is required. Temperature-adjusted, nonnumeric, specimen-ambiguous/conflicting, unsupported-unit and out-of-range measurements do not establish usable evidence.

Ascertainment is encounter-wide, including results recorded later in the stay or after treatment. It is not restricted to the encounter-start day or a pretreatment draw. Separate ABG and VBG observation models use the primary fixed BMI spline and the shared setting-inclusive adjustment, standardized over all adjustment-complete primary patients. The endpoints indicate usable captured evidence, not every test ordered or drawn. No usable result does not establish no testing or physiological normality. Patients can have both specimens, so these probabilities are not mutually exclusive. In F1-main, panel C's gray band describes the joint total; component intervals appear in the landmark table.

2.2 Historical source-entry populations

A confirmed positive ABG, VBG, RESPFAIL, VENTSUPPORT or PREDISPOSITION source flag establishes the primary population, including encounters also represented through OBESITY. Obesity-only membership requires OBESITY and authenticated zeros in all five other families. Obesity-only indexes are excluded from scientific comparisons: removing the gas-related entry pathways selects against the evidence needed to assess the physiological outcomes. Their counts remain in the detailed audit flow for reconciliation; source discrepancies and descriptive profiles are confined to the audit. Unknown flags remain unknown. All-six-zero discrepancies require source investigation. RFS metadata are applied after the common index is selected.

The combined indexed cohort is a reconciliation reference. Non-OBESITY-only, overlapping and unknown-OBESITY/confirmed-non-OBESITY subgroups are descriptive. BMI ≥40 remains a descriptive stratum and is not equated with obesity-route membership. The nested BMI ≥30 secondary analysis retains the newly selected primary index and never promotes a later obese encounter. All three endpoint families are refitted within this secondary population. The overlapping all-OBESITY-route analysis and current-measurement analysis remain supporting checks.

2.3 Outcomes and adjustment

Arterial hypercapnia requires ordinary authenticated ABG $PaCO_2$ ≥45 mm Hg. Combined gas accepts ABG $PaCO_2$ ≥45 or VBG $PvCO_2$ ≥50 mm Hg; both specimens are not required. No pH restriction is applied to the established primary outcomes. Strict validity limits are 5<arterial<200 and 10<venous<250 mm Hg, with reviewed units. Available-evidence negatives do not establish that an unmeasured specimen is normal. Unusable or unavailable outcomes remain unknown.

The ICD composite comprises J96.02, J96.12, J96.22, J96.92 and E66.2, with captured absence negative only under complete diagnosis capture. E66.2 introduces obesity-related ascertainment. Index-calendar-day gas sensitivities use recorded-result date equal to encounter-start date, retaining the index. This export does not authenticate collection times.

Adjustment includes linear age, inherited sex, combined race/ethnicity and geographic-region coding, and index encounter setting (EMER emergency as reference; IMP inpatient) as a categorical main effect in every fitted model. No BMI-by-setting interaction is fitted. Standardization retains each patient's observed setting. Completeness means nonmissing values for inherited analysis variables, including retained Unknown categories. The White reference category includes non-Hispanic or unknown ethnicity; inherited race-category precedence is unchanged. Raw race × ethnicity cross-tabs are shown only when authenticated.

2.4 Models, probabilities and uncertainty

Each main population uses type-7 BMI quantiles .05/.35/.65/.95 from its own adjustment-complete population: two interior knots, two boundaries and three natural-spline basis columns. The basis remains fixed across its binary and joint endpoint families and predictions; primary timing and matched checks retain the primary basis. The primary current-measurement sensitivity reuses this basis. The overlapping all-OBESITY supporting analysis uses its own basis and is reported in the audit. Obesity-only and obesity-only current-measurement models are not registered. Constant outcomes, duplicate knots, separation, rank and convergence failures in planned models remain explicit without estimator substitution.

Gas curves standardize over each model's outcome-evaluable population; ICD uses known capture within group. Observation curves describe usable gas-evidence availability. Complete diagnosis capture is reported as a source fact, not as ICD positivity or a constant availability curve. No physiological outcomes are projected to untested patients. Contrasts over supported five-unit intervals are audit summaries, including 40 versus 45; the full supported BMI range organizes the presentation. Local counts and marginal covariate coverage expose sparsity even inside the global BMI range; these checks do not establish joint covariate support.

Pointwise 95% intervals use full coefficient covariance and condition on the empirical reference and fixed spline basis. They omit uncertainty from selection, measurement error and unmeasured confounding. Apparent calibration is in-sample; fitted/observed mean agreement is not independent validation.

2.5 Secondary scenario and supporting checks

The named secondary scenario subtracts exactly one BMI unit when observed BMI >30; BMI≤30 is unchanged. There is no floor or reselection. Paired predictions hold index, RFS membership and other covariates fixed. Reported quantities are observed and shifted mean probabilities, their signed difference and PIF=1−p_shifted/p_observed. Patients with BMI ≤30 remain in the entire model reference denominator and baseline modeled burden; their scenario difference is exactly zero. Signed original-BMI-band contributions and their full covariance reproduce the total difference and variance. Marginal curve-density integration is not used.

Same-sample linear-BMI checks and matched ICD/gas analyses remain registered. A limited current-measurement sensitivity uses the already-selected BMI when reported at the index or calculated from height and weight both authenticated to that encounter. Full-cohort MAR/MNAR projections and IPW are excluded. The report is rendered from saved aggregates without clinical reads or refitting.

2.6 Same-day physiological components

The physiological component analysis uses any recorded calendar date within the existing index encounter with both usable ordinary arterial PCO_2 and arterial pH. It is **same recorded-day co-occurrence; not verified same-sample pairing**. The export dates when results were recorded; it does not identify a shared blood sample, collection time or elapsed 24-hour interval. Validity screens are $5<PCO_2<200$ mm Hg and $6.5<pH<7.8$, applied before display rounding.

The source endpoints use the same patients with at least one co-observed day: any $PaCO_2 \geq 45$ on such a day, and any such day also containing pH <7.35. Results may come from different samples. Elevated PCO_2 on one date and low pH on another cannot establish co-occurrence. Patients without a valid co-observed day have unknown component outcomes. Additional unusable or single-analyte records remain audited without erasing usable days. A negative describes eligible recorded evidence only. The relevant population's numeric BMI spline basis, specified adjustment and common component reference population are retained. A single unpenalized multinomial logistic spline assigns each patient to no documented hypercapnia, hypercapnia without documented same-day acidemia, or hypercapnia with same-day acidemia. Any eligible hypercapnia–acidemia day takes priority, including patients with other hypercapnic days without acidemia. The baseline outcome is no documented hypercapnia. All three probabilities are standardized over the same patients; full cross-category covariance supplies pointwise intervals and comparisons of component changes. No additional PIF is calculated.

The full observed BMI range is the main relationship, with sparse regions flagged. BMI 40–45 is a prespecified numerical summary, not a biological cutoff or fitting restriction. Every supported five-unit interval is also reported. These are descriptive pointwise contrasts without multiplicity adjustment; separate significance counts do not establish differences in associations.

2.7 Diagnostic components

The diagnostic component analysis partitions the inherited five-code ICD composite on the same fixed indexes within each main population. It requires complete encounter diagnosis capture and nonmissing specified adjustment variables, including retained Unknown categories; it does not require blood gas evidence. J96.22 or J96.02 together with J96.12/E66.2 receives priority as **acute-on-chronic or concurrent acute/chronic coding**. Remaining J96.02 is acute; remaining J96.12/E66.2 is chronic coding/OHS; remaining J96.92 is unspecified acuity. No target code with complete capture is the reference category. Incomplete capture is unknown. OHS is grouped analytically with chronic coding and does not itself establish chronic respiratory failure.

One Firth mean-bias-reduced five-category multinomial spline reuses its population's numeric BMI basis, linear age, inherited sex, race/ethnicity and region, and categorical encounter-setting adjustment. All class probabilities are standardized over the same diagnosis-evaluable patients. The estimator maximizes the multinomial log likelihood plus one half of the log determinant of its Fisher information. It replaces a separated unpenalized fit without pooling demographic levels or changing categories. The inverse multinomial information at the bias-reduced estimate supplies first-order joint covariance, with the empirical reference and numeric spline held fixed. This finite-estimate method does not resolve sparse support, source selection or measurement bias. Full joint covariance supplies pointwise 95% intervals, supported five-unit changes and pairwise differences in those changes. These descriptive comparisons are not multiplicity adjusted. The fitted positive components sum to the joint composite; it need not equal the existing binary model's predictions. No component-specific PIF is calculated.

The unpenalized four-category sensitivity excludes only the final unspecified-only category. Mixed codes assigned to specified categories remain included. It retains its population's fixed basis and adjustment but uses its own reduced fitting/reference population. This is not a same-denominator estimator comparison. Failed positions remain explicit.

3 Results

3.1 Cohort construction and evidence availability

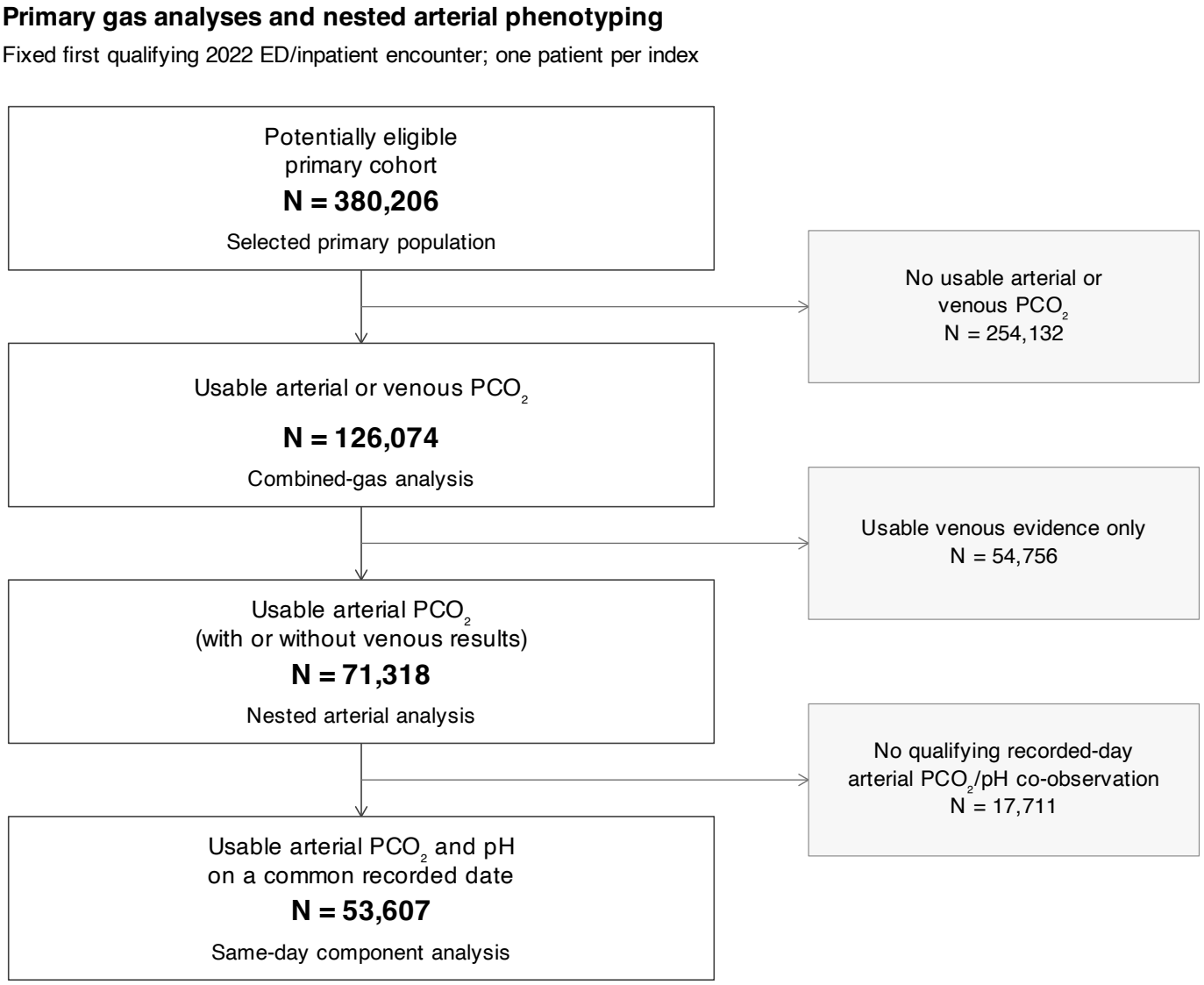


Figure 1: F0-primary-gas. Nested gas-evaluable populations at the fixed first qualifying 2022 emergency/inpatient encounter, one patient per index. The starting box is the existing primary source-entry cohort, not all TriNetX acute-care patients. Existing eligibility and index selection remain unchanged. Analysis samples retain the specified adjustment-completeness rules; any adjustment exclusions are listed separately from unavailable gas evidence. The arterial population includes patients with both arterial and venous results. Venous-only patients remain in the combined-gas analysis; patients without qualifying pH co-observation remain in the arterial analysis. No usable gas does not establish that no test was ordered or drawn. Normal and elevated PCO_2 are eligible; the last stage requires usable arterial pH, not acidemia. Usable arterial PCO_2 and pH must share at least one recorded date within that same index encounter; the date need not be its first calendar day. Same recorded-day co-occurrence; not verified same-sample pairing. Results may originate from different samples or treatment periods. Counts and evidence statuses come from the cohort ledger; unavailable recorded-day evidence is not zero.

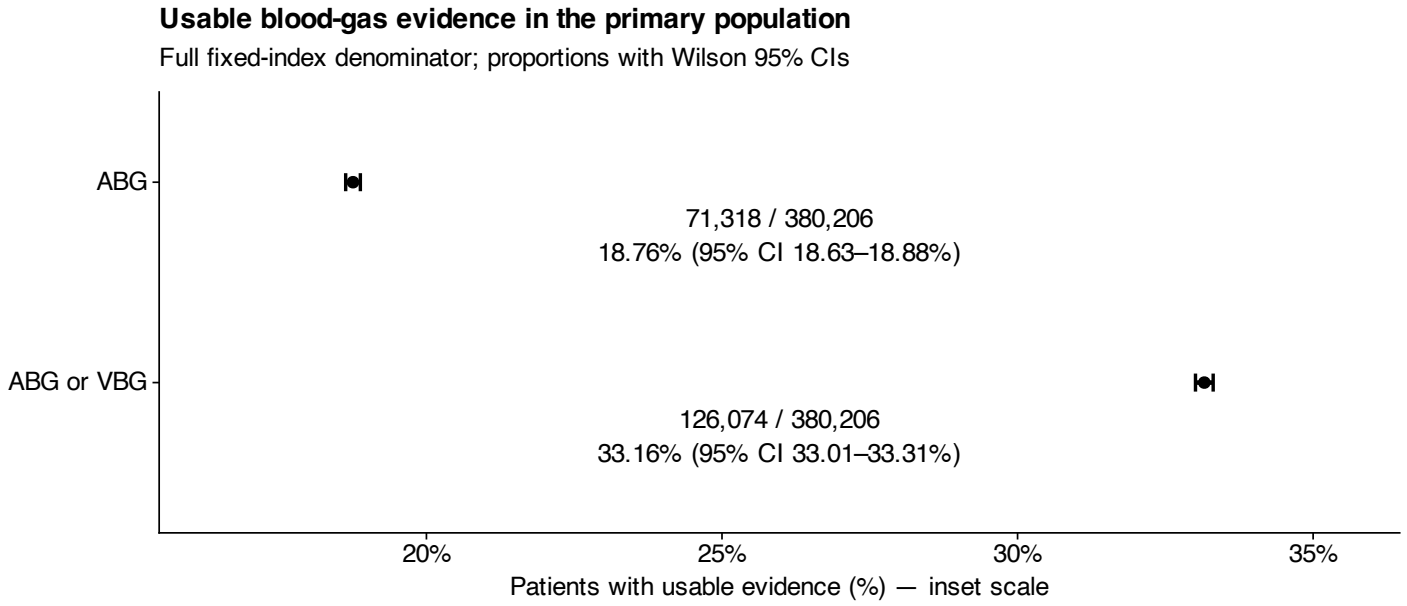


Figure 2: Encounter-wide usable gas evidence among all primary fixed indexes, before adjustment completeness. Labels give evaluable/full N, percentage and 95% Wilson CI; capped intervals use their exact endpoints on a labeled inset scale. ABG: arterial blood gas; VBG: venous blood gas. The combined row accepts either usable specimen; the two rows overlap and are not summed. These percentages describe evidence availability, not hypercapnia positivity. Primary includes non-OBESITY pathways and their OBESITY overlaps. Complete diagnosis capture, when present, is a source fact rather than ICD positivity and is reported separately. Obesity-only counts and residual-gas discrepancies are confined to the cohort flow and audit; they do not define a scientific comparator or weighting analysis.

Table 1: Fixed-index RFS partition

population id	n	complete	incomplete	denominator	status
primary	380206	380206	0	427079	assessed
obesity only	46873	46873	0	427079	assessed
unresolved	0	0	0	427079	assessed

Table 2: Encounter setting in the primary reference

level	population id	variable id	n	n reference	percent all	percent nonmissing	analysis population id
[MISSING]	primary::C	encounter type	0	380206	0	unavailable	primary
EMER	primary::C	encounter type	156509	380206	41.16426	41.16426	primary
IMP	primary::C	encounter type	223697	380206	58.83574	58.83574	primary
OTHER	primary::C	encounter type	0	380206	0	0	primary

label: Q; grain: patient; status: assessed.

Table 3: Nested secondary membership at the same index

population id	n records	n encounters	n patients	n	analysis population id
primary::Q	380206	380206	380206	380206	primary
primary bmi ge30::Q	175271	175271	175271	175271	primary bmi ge30

Known diagnosis capture: 380,206 / 380,206 primary indexes. Capture is not ICD positivity. All patient/outcome counts below use the common cohort ledger.

3.2 All valid BMI: primary

grain: patient; status: assessed; analysis population id: primary.

Table 4: Population and outcome denominators

population id	label	n records	n encounters	n patients	n
Q	Q	380206	380206	380206	380206
missing adjustment	missing adjustment	0	0	0	0
C	C	380206	380206	380206	380206
C icd	C icd	380206	380206	380206	380206
C abg encounter	C abg encounter	71318	71318	71318	71318
C any gas encounter	C any gas encounter	126074	126074	126074	126074

variable_id=age_at_encounter

n missing: 0; status: assessed; minimum: 18; q95: 84; q99: 88; n invalid: 0; analysis population id: primary.

Table 5: Participant profiles — panel 1/2

population id	n	n observed	n distinct	mean	sd	q01	q05
Q	380206	380206	82	59.30784	17.73305	20	26
C	380206	380206	82	59.30784	17.73305	20	26
C abg encounter	71318	71318	76	62.46516	16.10967	21	30

population id	n	n observed	n distinct	mean	sd	q01	q05
C any gas encounter	126074	126074	81	60.67596	17.03645	20	27

Table 6: Participant profiles — panel 2/2

population id	q25	median	q75	maximum
Q	47	62	73	106
C	47	62	73	106
C abg encounter	54	65	74	95
C any gas encounter	50	64	74	106

variable_id=bmi

n missing: 0; status: assessed; maximum: 89.8000030517578; n invalid: 0; analysis population id: primary.

Table 7: Participant profiles — panel 1/2

population id	n	n observed	n distinct	mean	sd	minimum	q01
Q	380206	380206	687	30.61148	8.580504	10.1	16.3
C	380206	380206	687	30.61148	8.580504	10.1	16.3
C abg encounter	71318	71318	622	29.5285	8.077181	10.2	16.1
C any gas encounter	126074	126074	653	29.6127	8.147867	10.1	16

Table 8: Participant profiles — panel 2/2

population id	q05	q25	median	q75	q95	q99
Q	19.2	24.4	29.1	35.6	46.6	55.3
C	19.2	24.4	29.1	35.6	46.6	55.3
C abg encounter	18.9	23.9	28.2	33.7	44.8	55
C any gas encounter	18.9	23.9	28.3	33.9	45	54.9

variable_id=sex

Unavailable throughout this group: mean, sd, minimum, q01, q05, q25, median, q75, q95, q99, maximum, n_invalid.

Table 9: Participant profiles

population id	n	n observed	n missing	n distinct	status	analysis population id
Q	380206	380206	0	2	assessed	primary
C	380206	380206	0	2	assessed	primary
C abg encounter	71318	71318	0	2	assessed	primary
C any gas encounter	126074	126074	0	2	assessed	primary

variable_id=race_ethnicity

Unavailable throughout this group: mean, sd, minimum, q01, q05, q25, median, q75, q95, q99, maximum, n_invalid.

Table 10: Participant profiles

population id	n	n observed	n missing	n distinct	status	analysis population id
Q	380206	380206	0	7	assessed	primary
C	380206	380206	0	7	assessed	primary
C abg encounter	71318	71318	0	7	assessed	primary
C any gas encounter	126074	126074	0	7	assessed	primary

variable_id=location

Unavailable throughout this group: mean, sd, minimum, q01, q05, q25, median, q75, q95, q99, maximum, n_invalid.

Table 11: Participant profiles

population id	n	n observed	n missing	n distinct	status	analysis population id
Q	380206	380206	0	4	assessed	primary
C	380206	380206	0	4	assessed	primary
C abg encounter	71318	71318	0	4	assessed	primary
C any gas encounter	126074	126074	0	4	assessed	primary

quantile type: 7; status: resolved; reference id: C; basis: splines::ns; intercept FALSE; 3 columns; analysis population id: primary; reference population id: C.

Table 12: Saved population-specific spline basis

basis id	knot probability	knot	kind
6094fd99aa4f5725f0a36afa82b83866f82625b4c83f87058d2bc53eafb813ec	0.05	19.2	boundary
6094fd99aa4f5725f0a36afa82b83866f82625b4c83f87058d2bc53eafb813ec	0.35	26.2	interior
6094fd99aa4f5725f0a36afa82b83866f82625b4c83f87058d2bc53eafb813ec	0.65	32.6	interior
6094fd99aa4f5725f0a36afa82b83866f82625b4c83f87058d2bc53eafb813ec	0.95	46.6	boundary

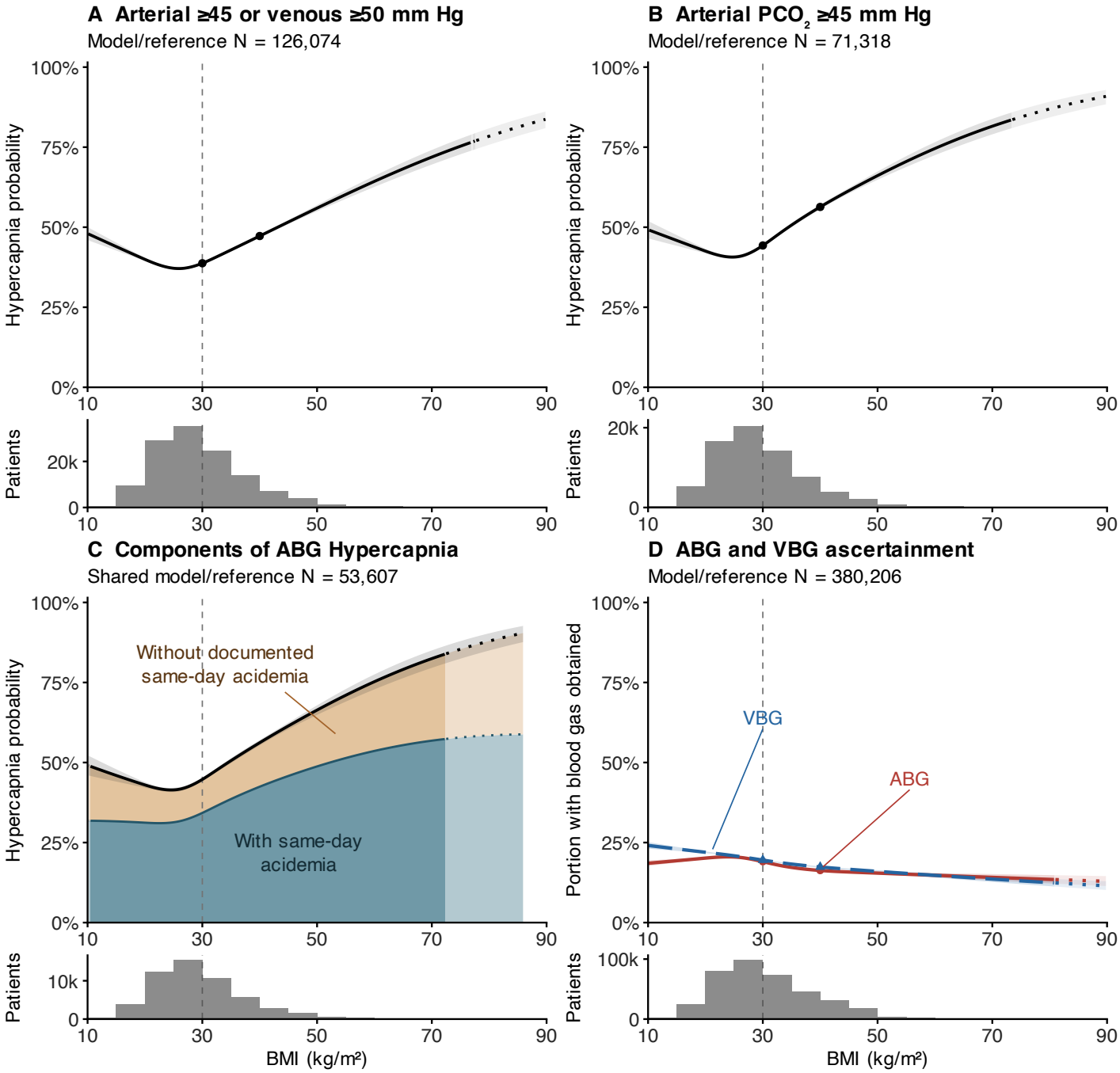
3.2.1 Overall documented hypercapnia

Arterial PCO_2 ≥45 mm Hg: 32,990 documented-positive outcomes among 71,318 model patients; observed positivity 46.26% (95% CI 45.89–46.62%) (Wilson interval). Model status: successful.

Arterial ≥45 or venous ≥50 mm Hg: 51,700 documented-positive outcomes among 126,074 model patients; observed positivity 41.01% (95% CI 40.74–41.28%) (Wilson interval). Model status: successful.

Documented ICD composite: 16,063 documented-positive outcomes among 380,206 model patients; observed positivity 4.225% (95% CI 4.161–4.289%) (Wilson interval). Model status: successful.

Hypercapnia across the range of BMIs



Landmarks: standardized probability, % (pointwise 95% CI)

Outcome / evidence	BMI 30 kg/m²	BMI 40 kg/m²
Combined-gas hypercapnia	38.7% (38.4–39.1)	47.2% (46.7–47.8)
Arterial hypercapnia	44.3% (43.8–44.8)	56.3% (55.6–57.1)
With same-day acidemia	34.3% (33.7–34.8)	42.4% (41.6–43.3)
Without documented same-day acidemia	10.5% (10.1–10.8)	13.7% (13.2–14.4)
ABG ascertainment	19.0% (18.9–19.2)	16.3% (16.1–16.5)
VBG ascertainment	19.5% (19.4–19.7)	17.4% (17.2–17.6)

95% CI: gray bands (A–C total), red/blue (D); component intervals in table.
Dotted/faded: sparse evidence. Frequency strips use each panel's reference.
Same recorded-day co-occurrence; not verified same-sample pairing.

F1-main. Hypercapnia across the range of BMIs in the selected primary historical source-entry population at fixed first 2022 emergency/inpatient indexes. BMI: body mass index; ABG: arterial blood gas; VBG: venous blood gas; PCO_2: partial pressure of carbon dioxide; CI: confidence interval. A: arterial PCO_2 ≥45 or venous PCO_2 ≥50 mm Hg (model/reference N = 126,074). B: arterial PCO_2 ≥45 mm Hg (N = 71,318); B includes patients with both specimens; these samples overlap and must not be summed. C: joint multinomial components in patients with usable arterial PCO_2 and arterial pH on a common recorded date (N = 53,607). Blue height gives hypercapnia with documented same-day acidemia (pH <7.35); orange thickness gives hypercapnia without such a day. Any eligible day with both findings takes priority. The upper boundary and gray 95% interval describe their total, not the orange component alone. Different samples or treatment periods may contribute. Same recorded-day co-occurrence; not verified same-sample pairing. D: separate ABG (red, solid) and VBG (blue, long-dashed) models describe usable captured PCO_2 evidence across the full adjustment-complete primary reference (N = 380,206). Obtained denotes a captured, usable PCO_2 result; tests without usable captured results are not identified. Either or both specimens may be available. Normal and elevated results count, with no pH requirement, at any time within the fixed encounter. All models reuse the primary fixed three-degree-of-freedom BMI spline and adjust for age, sex, race/ethnicity, geographic region and encounter setting (EMER reference; IMP inpatient). A–C standardize over their own outcome-evaluable reference; D over all adjustment-complete primary patients. Shading and table intervals are pointwise model-based 95% confidence intervals conditional on the empirical reference and fixed basis, not intervals for a BMI 30–40 contrast. Frequency strips describe each panel's corresponding reference. The vertical line marks BMI 30 (kg/m²); dotted/faded segments indicate sparse or out-of-range support. Full saved curve ranges are retained. Historical selection, selective measurement, treatment timing and residual confounding limit interpretation; these results do not establish acute respiratory failure, compensation, causal effects or physiological outcomes in untested patients.

family: primary; window: encounter; model form: spline; source status: qualified; formula text: .outcome ~ bmi_ns1 + bmi_ns2 + bmi_ns3 + age_at_encounter + sex + race_ethnicity + location + encounter_type; basis id: 6094fd99aa4f5725f0a36afa82b83866f82625b4c83f87058d2bc53eafb813ec; adjustment set id: inherited_plus_encounter_setting_v1; weight specification: none; covariance method: model-based binomial; empirical reference fixed; n eligible: 380206; n excluded adjustment: 0; n parameters: 16; model rank: 16; status: successful; failure reason: ; warnings: ; analysis population id: primary; ascertainment window: encounter; population id: primary; source reason: unavailable; estimator: unavailable; separation detected: unavailable.

Table 13: Binary model status — panel 1/2

model id	outcome id	training population id	field	n excluded outcome	n model	n patients	n events
abg/encounter/spline	abg	C abg encounter	abg	308888	71318	71318	32990
any gas/encounter/spline	any gas	C any gas encounter	any gas	254132	126074	126074	51700
icd/encounter/spline	icd	C icd	icd	0	380206	380206	16063

Table 14: Binary model status — panel 2/2

model id	n nonevents	fit runtime
abg/encounter/spline	38328	0.405
any gas/encounter/spline	74374	0.687
icd/encounter/spline	364143	2.586

3.2.2 Same-day physiological components

Table 15: Recorded-day source status

module	status	reason	n primary complete	n coobserved	analysis population id
same day arterial cooccurrence v1	qualified	Recorded-day evidence assessed; zeros refer only to co-observed days	380206	53607	primary

Table 16: Patient selection, day restriction and acidemia; Wilson 95\% intervals

Stage	Patients	Events	Observed	Status
Original arterial-evaluable	71318	32990	46.26% (95% CI 45.89–46.62%)	assessed
Co-observed patients: original arterial outcome	53607	25044	46.72% (95% CI 46.3–47.14%)	qualified
Same patients: hypercapnia on co-observed days	53607	25001	46.64% (95% CI 46.22–47.06%)	qualified
Same patients: same-day hypercapnia and acidemia	53607	18801	35.07% (95% CI 34.67–35.48%)	qualified

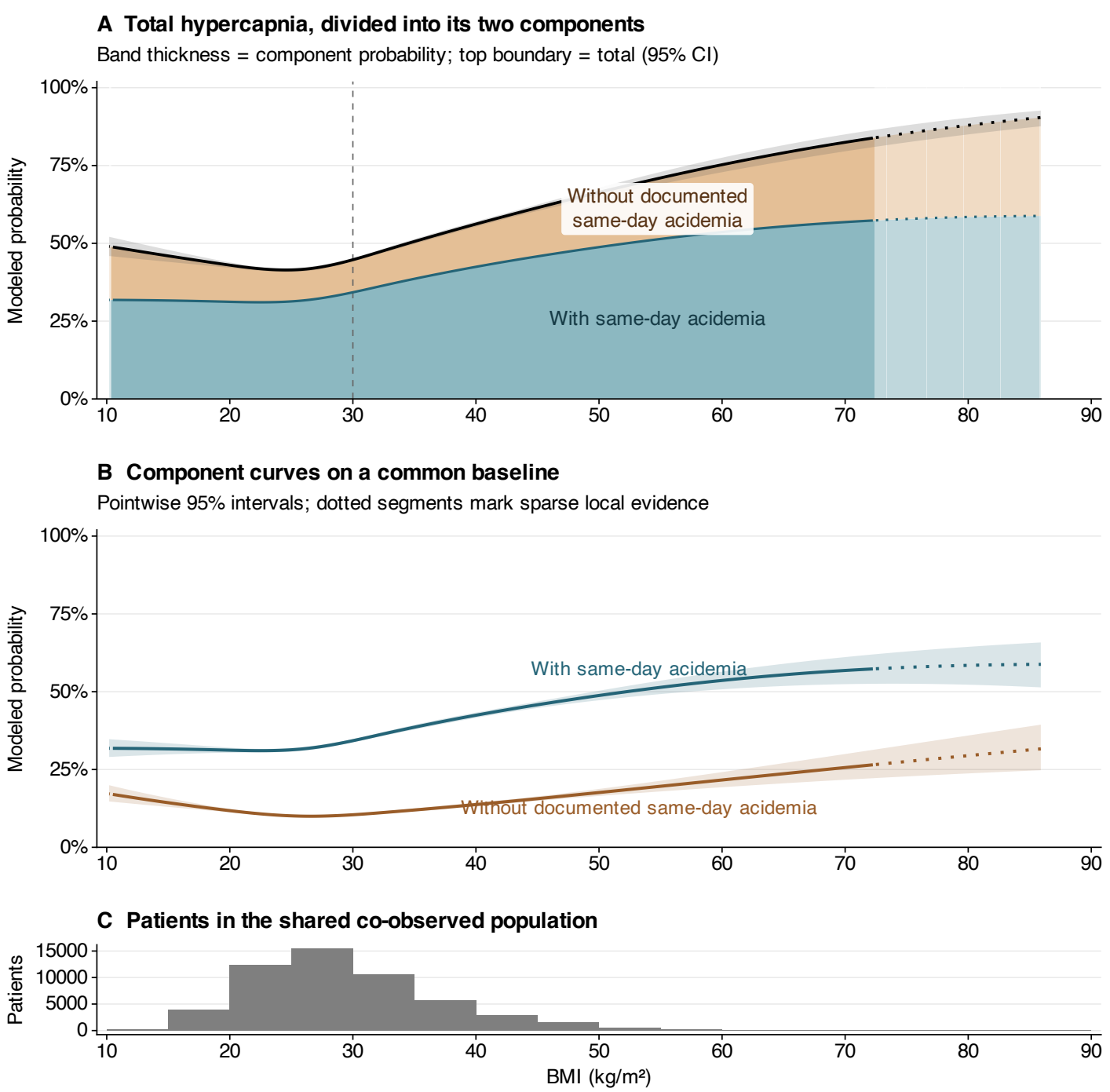


Figure 3: All valid BMI: primary. Same recorded-day co-occurrence; not verified same-sample pairing. Fixed-index patients with both arterial analytes share inherited plus categorical encounter-setting adjustment and their analysis population's fixed BMI spline. Any eligible day with $PaCO_2$ ≥45 mm Hg and pH <7.35 assigns the with-acidemia category; remaining hypercapnic patients have no documented same-day acidemia. Different samples may contribute. A: band thicknesses give component probabilities; the upper boundary is total hypercapnia. B: components share a baseline. Shading denotes pointwise 95% intervals using full joint covariance. C: shared-population BMI frequencies. Fading/dotted lines mark local windows below 30 patients or five events/non-events; marginal checks do not establish joint support. These recorded outcomes do not establish simultaneous physiology, acute respiratory failure or compensation. Selection, treatment timing and confounding remain limitations.

Table 17: Mutually exclusive same-day components

category id	n	N	status	definition	analysis population id
no hypercapnia	28606	53607	qualified	No documented hypercapnia on co-observed days	primary
without acidemia	6200	53607	qualified	Hypercapnia without documented same-day acidemia on any eligible day	primary
with acidemia	18801	53607	qualified	Any eligible day with hypercapnia and acidemia; takes priority over other days	primary

Same recorded-day co-occurrence; not verified same-sample pairing. Different samples and treatment periods may contribute. Requiring co-observation changes the available-evidence population; no inference about compensation or acute respiratory failure follows.

3.2.3 Diagnostic components

Table 18: Code-level diagnosis source status

module	status	reason	n primary complete	n complete capture	analysis population id
icd acuity v1	qualified	Complete encounter diagnosis capture; inherited five-code composite	380206	380206	primary

Table 19: Joint model estimability

model id	estimator	n model	status	failure reason
same day abg/encounter/multinomial spline	Unpenalized multinomial logit	53607	successful	
icd acuity/encounter/multinomial spline	firth mean bias reduced v1	380206	successful	
icd acuity specified/encounter/multinomial spline	maximum likelihood	379938	successful	

Table 20: Observed diagnostic components; Wilson 95\% intervals

Category	Patients	N	Observed	Status
No target diagnosis documented	364143	380206	95.78% (95% CI 95.71–95.84%)	qualified
Chronic coding / OHS	2198	380206	0.5781% (95% CI 0.5545–0.6027%)	qualified
Acute-on-chronic or concurrent coding	6258	380206	1.646% (95% CI 1.606–1.687%)	qualified
Acute coding	7339	380206	1.93% (95% CI 1.887–1.974%)	qualified
Unspecified acuity	268	380206	0.07049% (95% CI 0.06254–0.07944%)	qualified

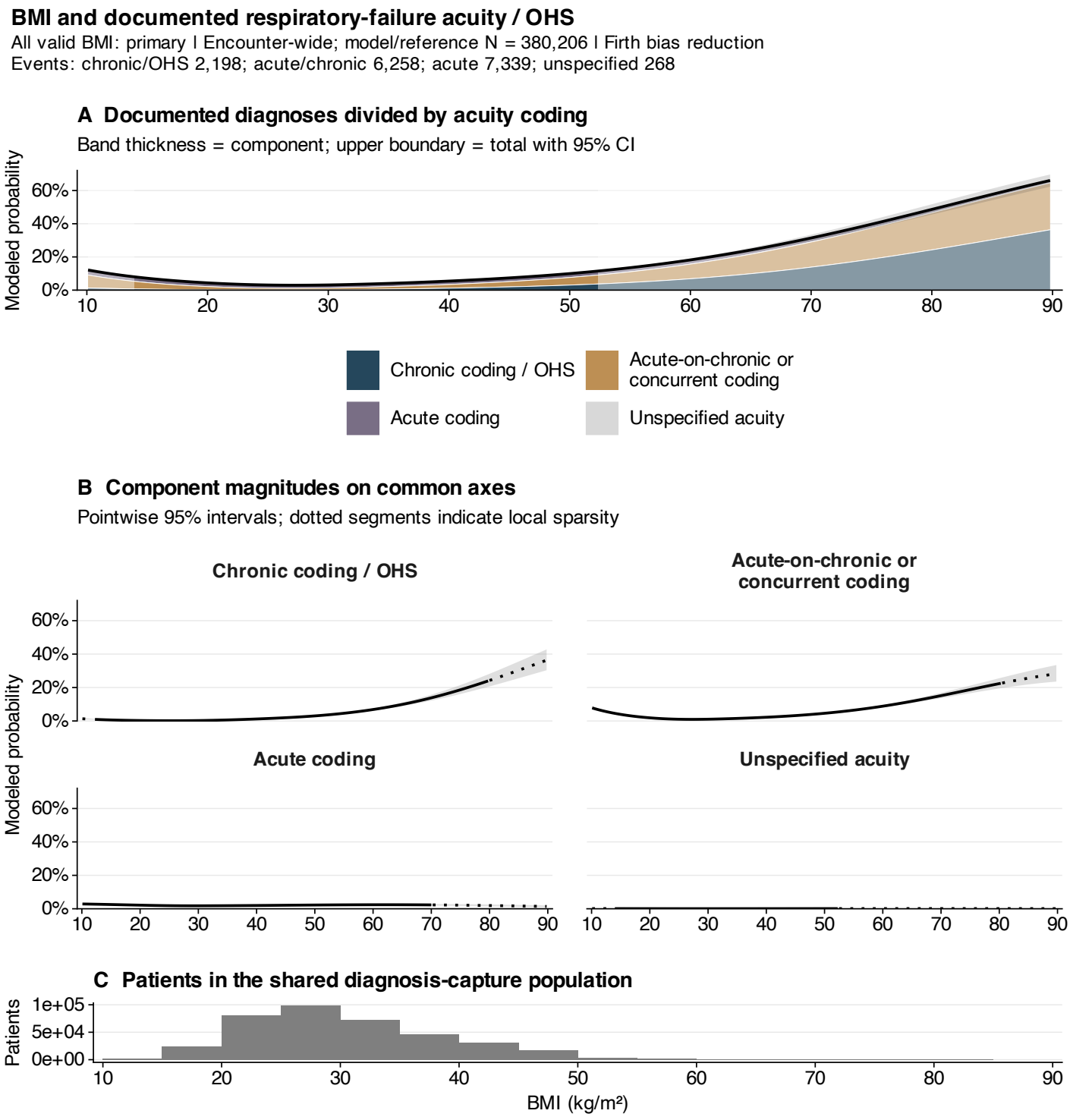


Figure 4: All valid BMI: primary. Mean-bias-reduced five-category multinomial spline; all diagnosis-evaluable patients in the stated population retained. Encounter-wide coding, no gas requirement; inherited plus categorical encounter-setting adjustment and the population-specific spline basis. Bands partition the composite; thickness gives component probability. Shading: pointwise 95% intervals from joint information covariance, conditional on the empirical reference and basis. Fading/dots: local sparsity; histogram N equals model/reference N. Acute-on-chronic includes explicit or concurrent acute/chronic coding. OHS (obesity hypoventilation syndrome) is analytically chronic; this does not establish chronic respiratory failure. Source selection and diagnostic incorporation limit physiological interpretation.

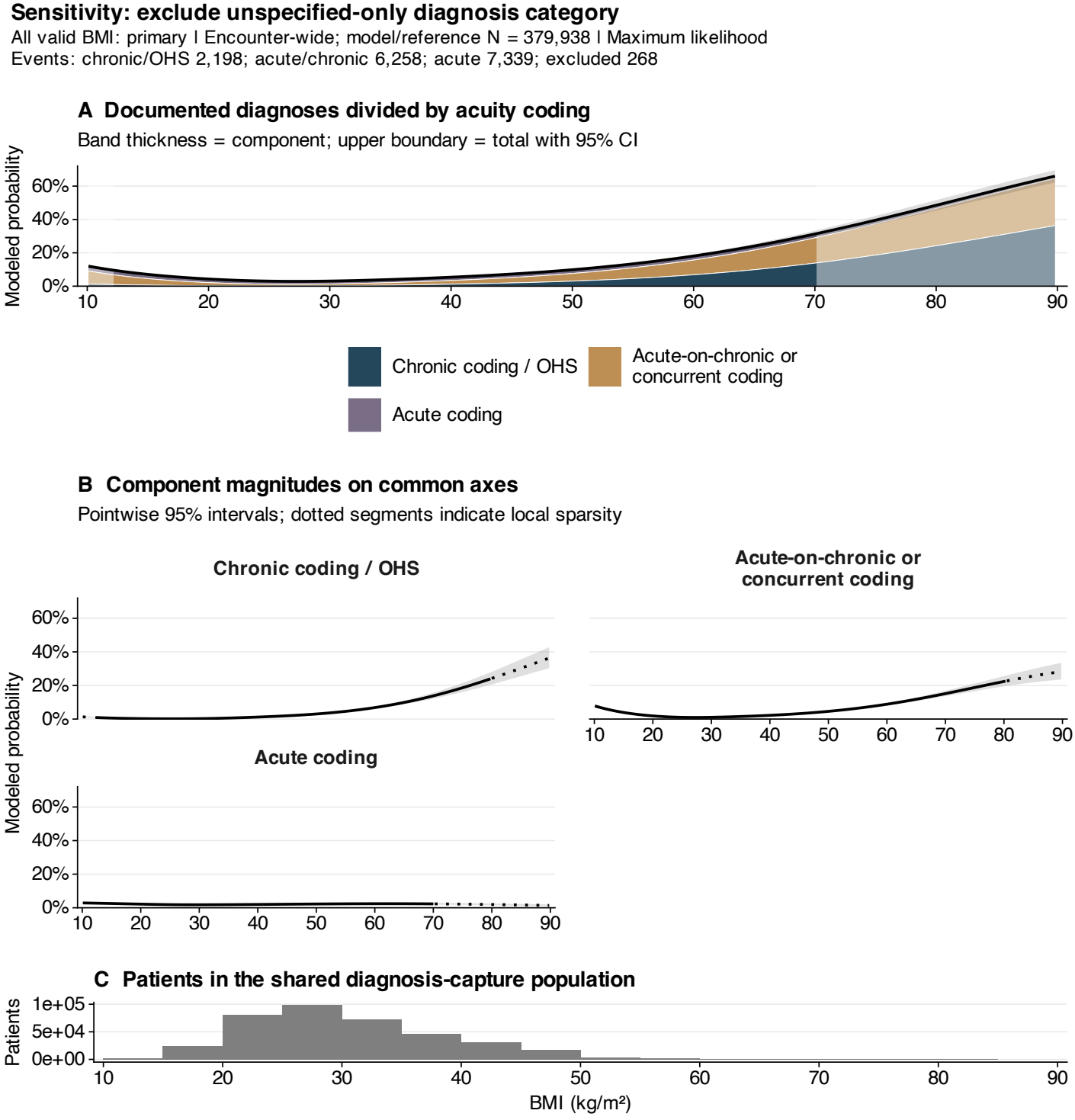


Figure 5: All valid BMI: primary. Unpenalized four-category sensitivity excluding unspecified-only encounters and standardizing over the remaining patients. Encounter-wide coding, no gas requirement; inherited plus categorical encounter-setting adjustment and the population-specific spline basis. Bands partition the composite; thickness gives component probability. Shading: pointwise 95% intervals from joint information covariance, conditional on the empirical reference and basis. Fading/dots: local sparsity; histogram N equals model/reference N. Acute-on-chronic includes explicit or concurrent acute/chronic coding. OHS (obesity hypoventilation syndrome) is analytically chronic; this does not establish chronic respiratory failure. Source selection and diagnostic incorporation limit physiological interpretation.

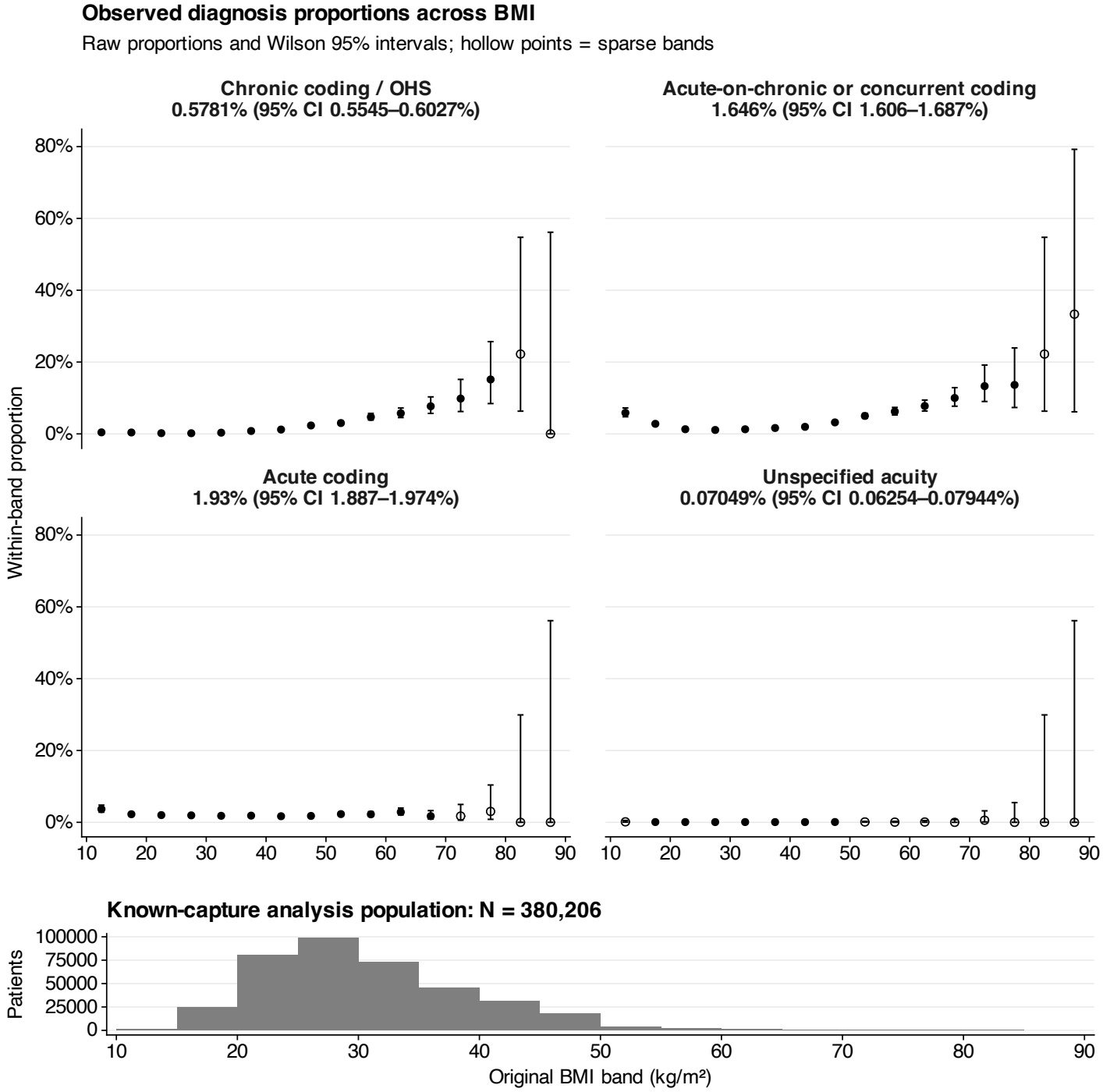


Figure 6: All valid BMI: primary. Observed encounter-wide diagnosis proportions within five-unit BMI bands of the covariate-complete analysis population with known capture; no gas requirement. These are unadjusted descriptions, not fitted curves. Error bars are Wilson 95% intervals; hollow points mark sparse patient/event/non-event counts. The histogram shows each band denominator, shared by every category. Empty bands have no estimate. OHS is grouped with chronic coding analytically; codes do not establish physiology. Overall percentages and their Wilson intervals are printed in each panel; no-target coding is the complement of the four displayed categories.

Diagnostic codes describe documentation, not acidemia or compensation. OHS is grouped analytically with chronic coding; diagnostic incorporation and source entry can affect its BMI association. Complete component contrast tables and their joint covariance are retained in the audit.

3.2.4 Modeled burden under the one-unit BMI scenario

Maximum decrement: 1.

Table 21: Paired model-standardized scenario — panel 1/2

Model	Reference	Model N	Reference N	Events	Patients shifted	Mean decrement	Eligible (%)
abg/encounter/spline	C abg encounter	71318	71318	32990	28616	0.4012451	40.12
any gas/encounter/ spline	C any gas encounter	126074	126074	51700	51345	0.4072608	40.73
icd/encounter/spline	C icd	380206	380206	16063	173099	0.4552769	45.53

Table 22: Paired model-standardized scenario — panel 2/2

Model	Observed probability % (95% CI)	Shifted probability % (95% CI)	Difference per 1000 (95% CI)	PIF % (95% CI)
abg/encounter/spline	46.26 (45.9, 46.62)	45.79 (45.43, 46.15)	4.675 (4.407, 4.943)	1.011 (0.952, 1.069)
any gas/encounter/ spline	41.01 (40.74, 41.28)	40.67 (40.4, 40.93)	3.418 (3.217, 3.619)	0.8334 (0.7842, 0.8826)
icd/encounter/spline	4.225 (4.162, 4.288)	4.104 (4.043, 4.166)	1.207 (1.152, 1.262)	2.858 (2.733, 2.982)

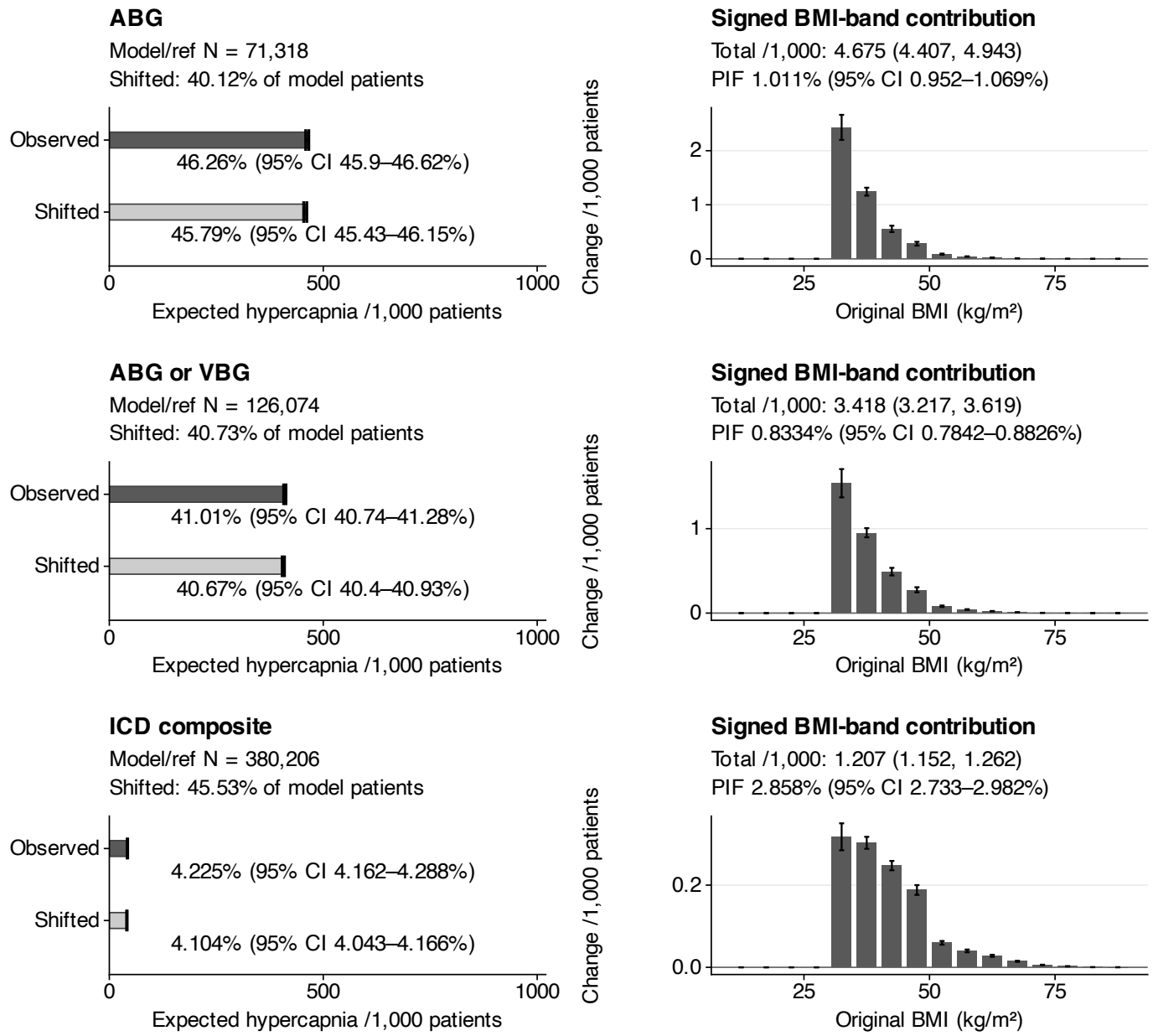


Figure 7: Analysis population: All valid BMI: primary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F2. Paired scenario: One-unit BMI reduction for observed BMI >30; no floor. ABG: arterial PCO_2 ≥45 mm Hg; ABG or VBG: arterial ≥45 or venous ≥50 mm Hg; ICD: five-code composite including E66.2. Left: observed and shifted standardized probabilities with model-based 95% CIs and printed limits. Left axes share the full 0–1,000 reference-patient scale; printed probabilities and limits retain small differences. N identifies the common model/reference population; shifted percentage is a descriptive cohort fraction. Right: signed original-BMI-band contributions per 1,000 reference patients, with 95% CIs. Patients at BMI ≤30 remain in baseline burden and contribute exactly zero scenario difference; the eligible share is printed. Total-difference/PIF limits are printed; contribution covariance reproduces total variance. Right axes vary by outcome and retain zero and all limits. Gas samples are outcome-evaluable; ICD uses known capture. Adjustment, source-entry membership and covariates are fixed. PIF is the model-standardized potential impact fraction. These conditional scenarios do not estimate a weight-loss treatment effect.

Baseline burden includes all reference patients, including those at BMI ≤30 who have zero scenario difference. PIF is relative to that entire modeled baseline burden. These conditional model changes are not estimates of preventable cases or treatment efficacy.

note: apparent in-sample diagnostic; no validation or causal claim; analysis population id: primary; population id: primary; ascertainment window: encounter; model form: spline.

Table 23: Binary apparent fit diagnostics

metric	model id	value	outcome id
log likelihood	abg/encounter/spline	-48299.91	abg
null deviance	abg/encounter/spline	98467.83	abg
residual deviance	abg/encounter/spline	96599.82	abg
AIC	abg/encounter/spline	96631.82	abg
Brier	abg/encounter/spline	0.2421657	abg
log loss	abg/encounter/spline	0.6772471	abg
iterations	abg/encounter/spline	4	abg
residual df	abg/encounter/spline	71302	abg
scaled condition number	abg/encounter/spline	9.59986	abg
events per parameter	abg/encounter/spline	2061.875	abg
nonevents per parameter	abg/encounter/spline	2395.5	abg
log likelihood	any gas/encounter/spline	-83716.76	any gas
null deviance	any gas/encounter/spline	170675.6	any gas
residual deviance	any gas/encounter/spline	167433.5	any gas
AIC	any gas/encounter/spline	167465.5	any gas
Brier	any gas/encounter/spline	0.2358417	any gas
log loss	any gas/encounter/spline	0.6640288	any gas
iterations	any gas/encounter/spline	4	any gas
residual df	any gas/encounter/spline	126058	any gas
scaled condition number	any gas/encounter/spline	8.778893	any gas
events per parameter	any gas/encounter/spline	3231.25	any gas
nonevents per parameter	any gas/encounter/spline	4648.375	any gas
log likelihood	icd/encounter/spline	-61560.58	icd
null deviance	icd/encounter/spline	133090.5	icd
residual deviance	icd/encounter/spline	123121.2	icd
AIC	icd/encounter/spline	123153.2	icd
Brier	icd/encounter/spline	0.03932209	icd
log loss	icd/encounter/spline	0.1619138	icd
iterations	icd/encounter/spline	7	icd
residual df	icd/encounter/spline	380190	icd
scaled condition number	icd/encounter/spline	7.927883	icd
events per parameter	icd/encounter/spline	1003.938	icd
nonevents per parameter	icd/encounter/spline	22758.94	icd

fit bmi max: 89.8000030517578; n observed outside: 0; n transformed outside: 0; pct transformed outside: 0; maximum observed gap: 0; maximum transformed gap: 0; n unseen categories: 0; analysis population id: primary; population id: primary.

Table 24: Curve and transformed prediction support — panel 1/2

model id	fit bmi min	n reference	reference id	policy id	method	reference population id	outcome id
abg/encounter/spline	10.2	71318	C abg encounter	bmi gt30 minus1 v1	observed	C abg encounter	abg
any gas/encounter/spline	10.1	126074	C any gas encounter	bmi gt30 minus1 v1	observed	C any gas encounter	any gas
icd/encounter/spline	10.1	380206	C icd	bmi gt30 minus1 v1	observed	C icd	icd

model id	fit bmi min	n reference	reference id	policy id	method	reference population id	outcome id
abg/index calendar day/spline	10.7	41785	C abg index calendar day	bmi gt30 minus1 v1	observed	C abg index calendar day	abg
any gas/index calendar day/spline	10.1	92324	C any gas index calendar day	bmi gt30 minus1 v1	observed	C any gas index calendar day	any gas
matched/abg/icd	10.2	71318	M abg	bmi gt30 minus1 v1	observed	M abg	icd
matched/abg/abg	10.2	71318	M abg	bmi gt30 minus1 v1	observed	M abg	abg
matched/any gas/icd	10.1	126074	M any gas	bmi gt30 minus1 v1	observed	M any gas	icd
matched/any gas/any gas	10.1	126074	M any gas	bmi gt30 minus1 v1	observed	M any gas	any gas
abg/encounter/spline	10.2	71318	C abg encounter	none; relationship	fixed BMI curve	C abg encounter	abg
abg/encounter/linear	10.2	71318	C abg encounter	none; relationship	fixed BMI curve	C abg encounter	abg
any gas/encounter/ spline	10.1	126074	C any gas encounter	none; relationship	fixed BMI curve	C any gas encounter	any gas
any gas/encounter/ linear	10.1	126074	C any gas encounter	none; relationship	fixed BMI curve	C any gas encounter	any gas
icd/encounter/spline	10.1	380206	C icd	none; relationship	fixed BMI curve	C icd	icd
icd/encounter/linear	10.1	380206	C icd	none; relationship	fixed BMI curve	C icd	icd
abg/index calendar day/spline	10.7	41785	C abg index calendar day	none; relationship	fixed BMI curve	C abg index calendar day	abg
abg/index calendar day/linear	10.7	41785	C abg index calendar day	none; relationship	fixed BMI curve	C abg index calendar day	abg
any gas/index calendar day/spline	10.1	92324	C any gas index calendar day	none; relationship	fixed BMI curve	C any gas index calendar day	any gas
any gas/index calendar day/linear	10.1	92324	C any gas index calendar day	none; relationship	fixed BMI curve	C any gas index calendar day	any gas
observation/abg/ encounter	10.1	380206	C	none; relationship	fixed BMI curve	C	abg
observation/vbg/ encounter	10.1	380206	C	none; relationship	fixed BMI curve	C	vbg
observation/any gas/ encounter	10.1	380206	C	none; relationship	fixed BMI curve	C	any gas
observation/abg/index calendar day	10.1	380206	C	none; relationship	fixed BMI curve	C	abg
observation/vbg/index calendar day	10.1	380206	C	none; relationship	fixed BMI curve	C	vbg
observation/any gas/ index calendar day	10.1	380206	C	none; relationship	fixed BMI curve	C	any gas
matched/abg/icd	10.2	71318	M abg	none; relationship	fixed BMI curve	M abg	icd
matched/abg/abg	10.2	71318	M abg	none; relationship	fixed BMI curve	M abg	abg
matched/any gas/icd	10.1	126074	M any gas	none; relationship	fixed BMI curve	M any gas	icd
matched/any gas/any gas	10.1	126074	M any gas	none; relationship	fixed BMI curve	M any gas	any gas

Table 25: Curve and transformed prediction support — panel 2/2

model id	ascertainment window	model form
abg/encounter/spline	encounter	spline
any gas/encounter/spline	encounter	spline
icd/encounter/spline	encounter	spline
abg/index calendar day/spline	index calendar day	spline
any gas/index calendar day/spline	index calendar day	spline
matched/abg/icd	encounter	spline
matched/abg/abg	encounter	spline
matched/any gas/icd	encounter	spline
matched/any gas/any gas	encounter	spline
abg/encounter/spline	encounter	spline
abg/encounter/linear	encounter	linear
any gas/encounter/spline	encounter	spline
any gas/encounter/linear	encounter	linear
icd/encounter/spline	encounter	spline
icd/encounter/linear	encounter	linear
abg/index calendar day/spline	index calendar day	spline
abg/index calendar day/linear	index calendar day	linear
any gas/index calendar day/spline	index calendar day	spline
any gas/index calendar day/linear	index calendar day	linear
observation/abg/encounter	encounter	spline
observation/vbg/encounter	encounter	spline
observation/any gas/encounter	encounter	spline
observation/abg/index calendar day	index calendar day	spline
observation/vbg/index calendar day	index calendar day	spline
observation/any gas/index calendar day	index calendar day	spline
matched/abg/icd	encounter	spline
matched/abg/abg	encounter	spline
matched/any gas/icd	encounter	spline
matched/any gas/any gas	encounter	spline

3.3 BMI 30 or higher: secondary

This secondary population is nested at the new all-BMI indexes; it is not the historical first-obese-encounter cohort. Its models are refitted with a separately fixed BMI basis.

grain: patient; status: assessed; analysis population id: primary_bmi_ge30.

Table 26: Population and outcome denominators

population id	label	n records	n encounters	n patients	n
Q	Q	175271	175271	175271	175271
missing adjustment	missing adjustment	0	0	0	0
C	C	175271	175271	175271	175271
C icd	C icd	175271	175271	175271	175271
C abg encounter	C abg encounter	29021	29021	29021	29021
C any gas encounter	C any gas encounter	52047	52047	52047	52047

variable_id=age_at_encounter

n missing: 0; status: assessed; minimum: 18; q95: 82; q99: 87; n invalid: 0; analysis population id: primary_bmi_ge30.

Table 27: Participant profiles — panel 1/2

population id	n	n observed	n distinct	mean	sd	q01	q05
Q	175271	175271	77	57.43002	16.77872	21	27
C	175271	175271	77	57.43002	16.77872	21	27
C abg encounter	29021	29021	75	61.5281	14.93395	23	32
C any gas encounter	52047	52047	76	59.70277	15.68799	22	30

Table 28: Participant profiles — panel 2/2

population id	q25	median	q75	maximum
Q	45	60	71	96
C	45	60	71	96
C abg encounter	53	64	73	92
C any gas encounter	50	62	72	96

variable_id=bmi

n missing: 0; status: assessed; minimum: 30; q01: 30; maximum: 89.8000030517578; n invalid: 0; analysis population id: primary_bmi_ge30.

Table 29: Participant profiles — panel 1/2

population id	n	n observed	n distinct	mean	sd	q05	q25
Q	175271	175271	488	37.952	6.700435	30.4	32.7
C	175271	175271	488	37.952	6.700435	30.4	32.7
C abg encounter	29021	29021	436	37.10515	6.704827	30.3	32.2
C any gas encounter	52047	52047	458	37.20281	6.662708	30.4	32.3

Table 30: Participant profiles — panel 2/2

population id	median	q75	q95	q99
Q	36.3	41.6	49.7	60.5
C	36.3	41.6	49.7	60.5
C abg encounter	35.2	40	49.8	61.4
C any gas encounter	35.3	40.2	49.7	60.8

variable_id=sex

Unavailable throughout this group: mean, sd, minimum, q01, q05, q25, median, q75, q95, q99, maximum, n_invalid.

Table 31: Participant profiles

population id	n	n observed	n missing	n distinct	status	analysis population id
Q	175271	175271	0	2	assessed	primary bmi ge30
C	175271	175271	0	2	assessed	primary bmi ge30
C abg encounter	29021	29021	0	2	assessed	primary bmi ge30
C any gas encounter	52047	52047	0	2	assessed	primary bmi ge30

variable_id=race_ethnicity

Unavailable throughout this group: mean, sd, minimum, q01, q05, q25, median, q75, q95, q99, maximum, n_invalid.

Table 32: Participant profiles

population id	n	n observed	n missing	n distinct	status	analysis population id
Q	175271	175271	0	7	assessed	primary bmi ge30
C	175271	175271	0	7	assessed	primary bmi ge30

population id	n	n observed	n missing	n distinct	status	analysis population id
C abg encounter	29021	29021	0	7	assessed	primary bmi ge30
C any gas encounter	52047	52047	0	7	assessed	primary bmi ge30

variable_id=location

Unavailable throughout this group: mean, sd, minimum, q01, q05, q25, median, q75, q95, q99, maximum, n_invalid.

Table 33: Participant profiles

population id	n	n observed	n missing	n distinct	status	analysis population id
Q	175271	175271	0	4	assessed	primary bmi ge30
C	175271	175271	0	4	assessed	primary bmi ge30
C abg encounter	29021	29021	0	4	assessed	primary bmi ge30
C any gas encounter	52047	52047	0	4	assessed	primary bmi ge30

quantile type: 7; status: resolved; reference id: C; basis: splines::ns; intercept FALSE; 3 columns; analysis population id: primary_bmi_ge30; reference population id: C.

Table 34: Saved population-specific spline basis

basis id	knot probability	knot	kind
50305a6ebd72a9e60f5e70f90207af5485824e3dc486048dab60919df1272f44	0.05	30.4	boundary
50305a6ebd72a9e60f5e70f90207af5485824e3dc486048dab60919df1272f44	0.35	34	interior
50305a6ebd72a9e60f5e70f90207af5485824e3dc486048dab60919df1272f44	0.65	39.3	interior
50305a6ebd72a9e60f5e70f90207af5485824e3dc486048dab60919df1272f44	0.95	49.7	boundary

3.3.1 Overall documented hypercapnia

Arterial PCO_2 ≥45 mm Hg: 15,209 documented-positive outcomes among 29,021 model patients; observed positivity 52.41% (95% CI 51.83–52.98%) (Wilson interval). Model status: successful.

Arterial ≥45 or venous ≥50 mm Hg: 23,179 documented-positive outcomes among 52,047 model patients; observed positivity 44.53% (95% CI 44.11–44.96%) (Wilson interval). Model status: successful.

Documented ICD composite: 8,438 documented-positive outcomes among 175,271 model patients; observed positivity 4.814% (95% CI 4.715–4.915%) (Wilson interval). Model status: successful.

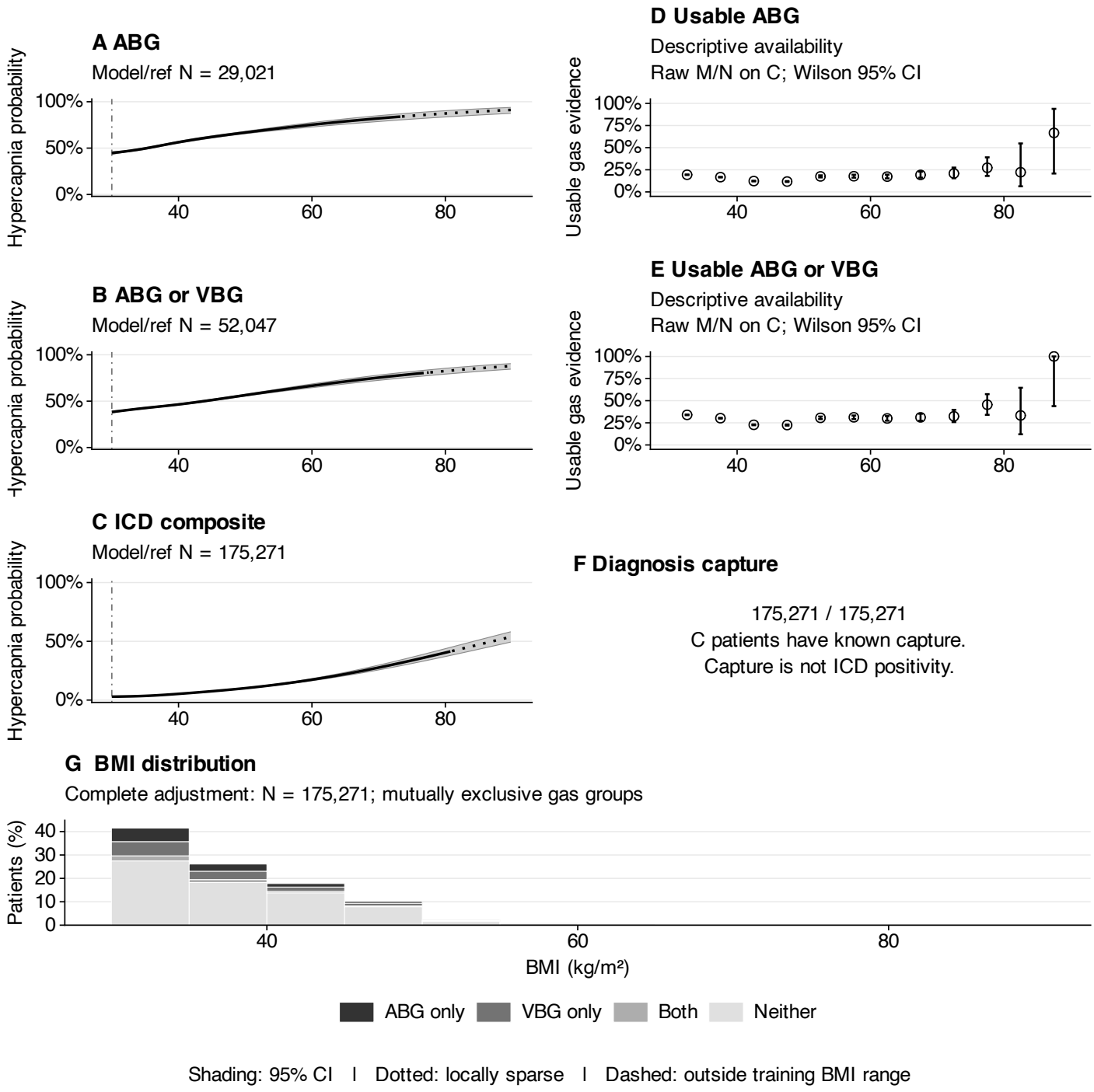


Figure 8: Analysis population: BMI ≥30: secondary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F1. BMI–hypercapnia relationships at the first eligible adult 2022 acute-care encounter. ABG denotes $PaCO_2$ ≥45 mm Hg; ABG or VBG denotes $PaCO_2$ ≥45 or $PvCO_2$ ≥50 mm Hg. ICD denotes the five-code composite, including E66.2. The vertical dot-dash line marks BMI 30, the scenario eligibility boundary and this secondary population's BMI cutoff. Gas curves standardize over outcome-evaluable patients; ICD uses known diagnosis capture. Model/ref N is the common model and reference count. Availability and BMI frequency use all adjustment-complete patients (C); the four frequency groups are mutually exclusive. Adjustment: age, sex, race/ethnicity, geographic region and encounter setting (EMER reference; IMP inpatient) with a fixed natural BMI spline. Shading and fine boundaries show pointwise model-based 95% CIs; dotted segments are locally sparse and dashed segments exceed training BMI support. Open availability points use Wilson intervals when no model is estimable. Known capture is distinct from ICD positivity. ABG/VBG: arterial/venous blood gas. Conditional model uncertainty excludes selection and measurement assumptions; no population transport or causal effect is inferred.

family: supporting; window: encounter; model form: spline; source status: qualified; formula text: .outcome ~ bmi_ns1 + bmi_ns2 + bmi_ns3 + age_at_encounter + sex + race_ethnicity + location + encounter_type; basis id: 50305a6ebd72a9e60f5e70f90207af5485824e3dc486048dab60919df1272f44; adjustment set id: inherited_plus_encounter_setting_v1; weight specification: none; covariance method: model-based binomial; empirical reference fixed; n eligible: 175271; n excluded adjustment: 0; n parameters: 16; model rank: 16; status: successful; failure reason: ; warnings: ; analysis population id: primary_bmi_ge30; ascertainment window: encounter; population id: primary_bmi_ge30; source reason: unavailable; estimator: unavailable; separation detected: unavailable.

Table 35: Binary model status — panel 1/2

model id	outcome id	training population id	field	n excluded outcome	n model	n patients	n events
abg/encounter/spline	abg	C abg encounter	abg	146250	29021	29021	15209
any gas/encounter/spline	any gas	C any gas encounter	any gas	123224	52047	52047	23179
icd/encounter/spline	icd	C icd	icd	0	175271	175271	8438

Table 36: Binary model status — panel 2/2

model id	n nonevents	fit runtime
abg/encounter/spline	13812	0.158
any gas/encounter/spline	28868	0.276
icd/encounter/spline	166833	1.075

3.3.2 Same-day physiological components

Table 37: Recorded-day source status

module	status	reason	n primary complete	n coobserved	analysis population id
same day arterial cooccurrence v1	qualified	Recorded-day evidence assessed; zeros refer only to co-observed days	175271	21748	primary bmi ge30

Table 38: Patient selection, day restriction and acidemia; Wilson 95\% intervals

Stage	Patients	Events	Observed	Status
Original arterial-evaluable	29021	15209	52.41% (95% CI 51.83–52.98%)	assessed
Co-observed patients: original arterial outcome	21748	11442	52.61% (95% CI 51.95–53.27%)	qualified
Same patients: hypercapnia on co-observed days	21748	11423	52.52% (95% CI 51.86–53.19%)	qualified
Same patients: same-day hypercapnia and acidemia	21748	8639	39.72% (95% CI 39.07–40.38%)	qualified

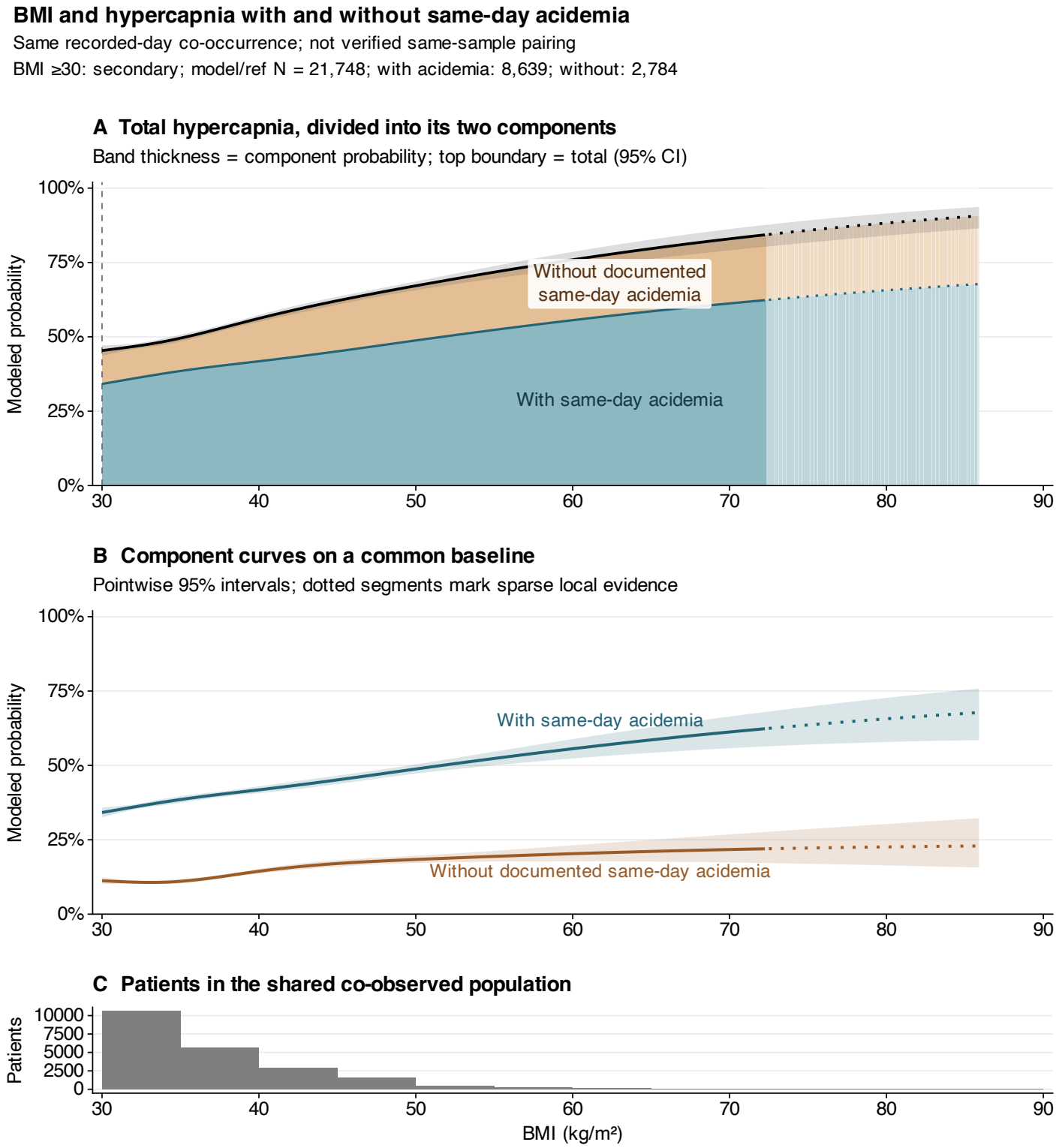


Figure 9: BMI ≥30: secondary. Same recorded-day co-occurrence; not verified same-sample pairing. Fixed-index patients with both arterial analytes share inherited plus categorical encounter-setting adjustment and their analysis population's fixed BMI spline. Any eligible day with $PaCO_2$ ≥45 mm Hg and pH <7.35 assigns the with-acidemia category; remaining hypercapnic patients have no documented same-day acidemia. Different samples may contribute. A: band thicknesses give component probabilities; the upper boundary is total hypercapnia. B: components share a baseline. Shading denotes pointwise 95% intervals using full joint covariance. C: shared-population BMI frequencies. Fading/dotted lines mark local windows below 30 patients or five events/non-events; marginal checks do not establish joint support. These recorded outcomes do not establish simultaneous physiology, acute respiratory failure or compensation. Selection, treatment timing and confounding remain limitations.

Table 39: Mutually exclusive same-day components

category id	n	N	status	definition	analysis population id
no hypercapnia	10325	21748	qualified	No documented hypercapnia on co-observed days	primary bmi ge30
without acidemia	2784	21748	qualified	Hypercapnia without documented same-day acidemia on any eligible day	primary bmi ge30
with acidemia	8639	21748	qualified	Any eligible day with hypercapnia and acidemia; takes priority over other days	primary bmi ge30

Same recorded-day co-occurrence; not verified same-sample pairing. Different samples and treatment periods may contribute. Requiring co-observation changes the available-evidence population; no inference about compensation or acute respiratory failure follows.

3.3.3 Diagnostic components

Table 40: Code-level diagnosis source status

module	status	reason	n primary complete	n complete capture	analysis population id
icd acuity v1	qualified	Complete encounter diagnosis capture; inherited five-code composite	175271	175271	primary bmi ge30

Table 41: Joint model estimability

model id	estimator	n model	status	failure reason
same day abg/encounter/multinomial spline	Unpenalized multinomial logit	21748	successful	
icd acuity/encounter/multinomial spline	firth mean bias reduced v1	175271	successful	
icd acuity specified/encounter/multinomial spline	maximum likelihood	175147	successful	

Table 42: Observed diagnostic components; Wilson 95\% intervals

Category	Patients	N	Observed	Status
No target diagnosis documented	166833	175271	95.19% (95% CI 95.08–95.28%)	qualified
Chronic coding / OHS	1752	175271	0.9996% (95% CI 0.9541–1.047%)	qualified
Acute-on-chronic or concurrent coding	3361	175271	1.918% (95% CI 1.854–1.983%)	qualified
Acute coding	3201	175271	1.826% (95% CI 1.765–1.89%)	qualified
Unspecified acuity	124	175271	0.07075% (95% CI 0.05935–0.08434%)	qualified

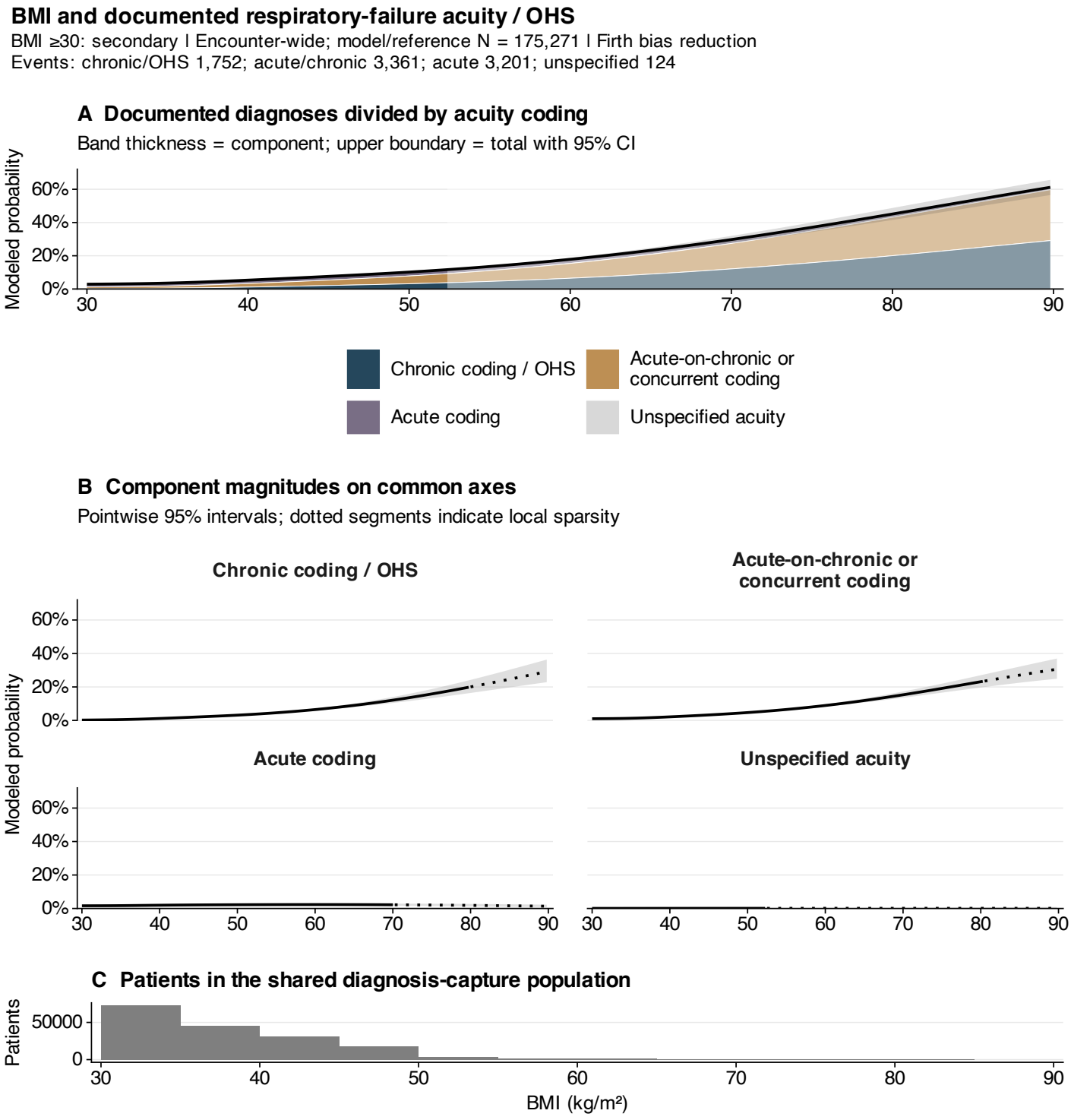


Figure 10: BMI ≥30: secondary. Mean-bias-reduced five-category multinomial spline; all diagnosis-evaluable patients in the stated population retained. Encounter-wide coding, no gas requirement; inherited plus categorical encounter-setting adjustment and the population-specific spline basis. Bands partition the composite; thickness gives component probability. Shading: pointwise 95% intervals from joint information covariance, conditional on the empirical reference and basis. Fading/dots: local sparsity; histogram N equals model/reference N. Acute-on-chronic includes explicit or concurrent acute/chronic coding. OHS (obesity hypoventilation syndrome) is analytically chronic; this does not establish chronic respiratory failure. Source selection and diagnostic incorporation limit physiological interpretation.

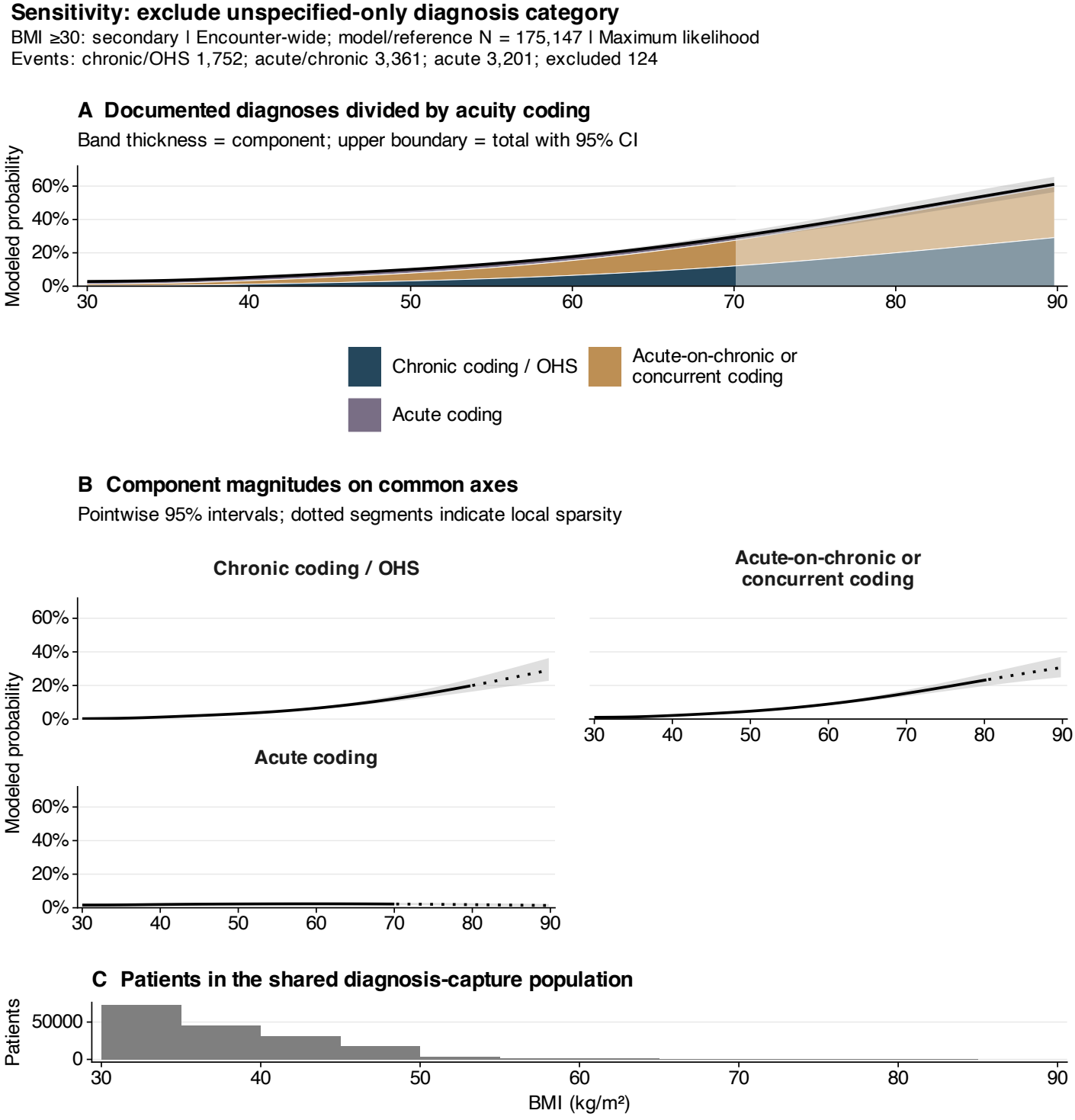


Figure 11: BMI ≥30: secondary. Unpenalized four-category sensitivity excluding unspecified-only encounters and standardizing over the remaining patients. Encounter-wide coding, no gas requirement; inherited plus categorical encounter-setting adjustment and the population-specific spline basis. Bands partition the composite; thickness gives component probability. Shading: pointwise 95% intervals from joint information covariance, conditional on the empirical reference and basis. Fading/dots: local sparsity; histogram N equals model/reference N. Acute-on-chronic includes explicit or concurrent acute/chronic coding. OHS (obesity hypoventilation syndrome) is analytically chronic; this does not establish chronic respiratory failure. Source selection and diagnostic incorporation limit physiological interpretation.

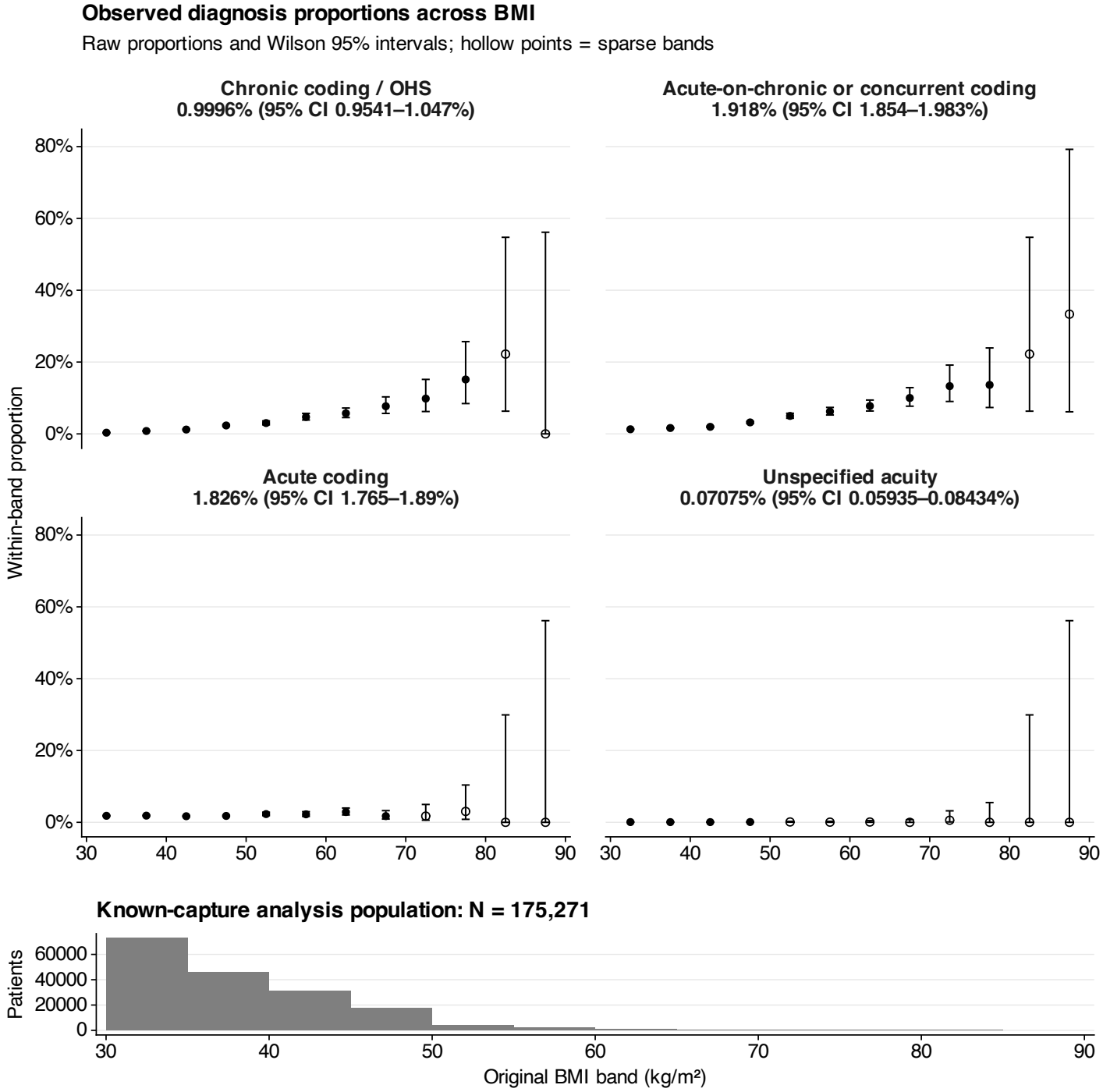


Figure 12: BMI ≥30: secondary. Observed encounter-wide diagnosis proportions within five-unit BMI bands of the covariate-complete analysis population with known capture; no gas requirement. These are unadjusted descriptions, not fitted curves. Error bars are Wilson 95% intervals; hollow points mark sparse patient/event/non-event counts. The histogram shows each band denominator, shared by every category. Empty bands have no estimate. OHS is grouped with chronic coding analytically; codes do not establish physiology. Overall percentages and their Wilson intervals are printed in each panel; no-target coding is the complement of the four displayed categories.

Diagnostic codes describe documentation, not acidemia or compensation. OHS is grouped analytically with chronic coding; diagnostic incorporation and source entry can affect its BMI association. Complete component contrast tables and their joint covariance are retained in the audit.

3.3.4 Modeled burden under the one-unit BMI scenario

Maximum decrement: 1.

Table 43: Paired model-standardized scenario — panel 1/2

Model	Reference	Model N	Reference N	Events	Patients shifted	Mean decrement	Eligible (%)
abg/encounter/spline	C abg encounter	29021	29021	15209	28616	0.9860446	98.6
any gas/encounter/ spline	C any gas encounter	52047	52047	23179	51345	0.9865122	98.65
icd/encounter/spline	C icd	175271	175271	8438	173099	0.9876078	98.76

Table 44: Paired model-standardized scenario — panel 2/2

Model	Observed probability % (95% CI)	Shifted probability % (95% CI)	Difference per 1000 (95% CI)	PIF % (95% CI)
abg/encounter/spline	52.41 (51.84, 52.97)	51.34 (50.74, 51.95)	10.63 (8.597, 12.67)	2.029 (1.639, 2.418)
any gas/encounter/ spline	44.53 (44.12, 44.95)	43.67 (43.23, 44.12)	8.603 (7.157, 10.05)	1.932 (1.606, 2.256)
icd/encounter/spline	4.814 (4.717, 4.913)	4.539 (4.444, 4.637)	2.751 (2.484, 3.019)	5.715 (5.169, 6.259)

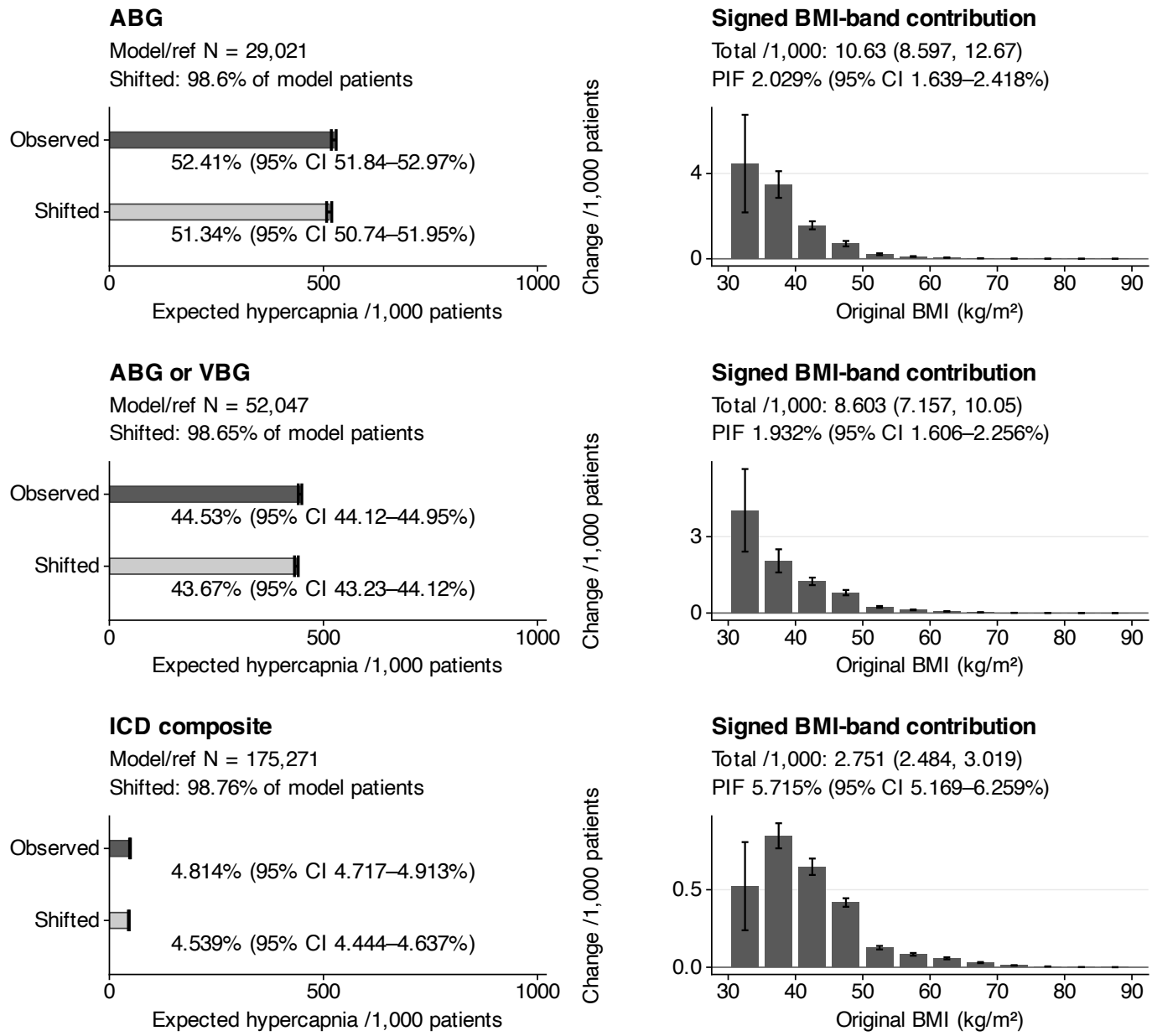


Figure 13: Analysis population: BMI ≥30: secondary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F2. Paired scenario: One-unit BMI reduction for observed BMI >30; no floor. ABG: arterial PCO_2 ≥45 mm Hg; ABG or VBG: arterial ≥45 or venous ≥50 mm Hg; ICD: five-code composite including E66.2. Left: observed and shifted standardized probabilities with model-based 95% CIs and printed limits. Left axes share the full 0–1,000 reference-patient scale; printed probabilities and limits retain small differences. N identifies the common model/reference population; shifted percentage is a descriptive cohort fraction. Right: signed original-BMI-band contributions per 1,000 reference patients, with 95% CIs. Patients at BMI ≤30 remain in baseline burden and contribute exactly zero scenario difference; the eligible share is printed. Total-difference/PIF limits are printed; contribution covariance reproduces total variance. Right axes vary by outcome and retain zero and all limits. Gas samples are outcome-evaluable; ICD uses known capture. Adjustment, source-entry membership and covariates are fixed. PIF is the model-standardized potential impact fraction. These conditional scenarios do not estimate a weight-loss treatment effect.

Baseline burden includes all reference patients, including those at BMI ≤30 who have zero scenario difference. PIF is relative to that entire modeled baseline burden. These conditional model changes are not estimates of preventable cases or treatment efficacy.

note: apparent in-sample diagnostic; no validation or causal claim; analysis population id: primary_bmi_ge30; population id: primary_bmi_ge30; ascertainment window: encounter; model form: spline.

Table 45: Binary apparent fit diagnostics

metric	model id	value	outcome id
log likelihood	abg/encounter/spline	-19670.08	abg
null deviance	abg/encounter/spline	40164.37	abg
residual deviance	abg/encounter/spline	39340.17	abg
AIC	abg/encounter/spline	39372.17	abg
Brier	abg/encounter/spline	0.2424679	abg
log loss	abg/encounter/spline	0.6777879	abg
iterations	abg/encounter/spline	4	abg
residual df	abg/encounter/spline	29005	abg
scaled condition number	abg/encounter/spline	10.55598	abg
events per parameter	abg/encounter/spline	950.5625	abg
nonevents per parameter	abg/encounter/spline	863.25	abg
log likelihood	any gas/encounter/spline	-34715.66	any gas
null deviance	any gas/encounter/spline	71529.38	any gas
residual deviance	any gas/encounter/spline	69431.32	any gas
AIC	any gas/encounter/spline	69463.32	any gas
Brier	any gas/encounter/spline	0.2373299	any gas
log loss	any gas/encounter/spline	0.667006	any gas
iterations	any gas/encounter/spline	4	any gas
residual df	any gas/encounter/spline	52031	any gas
scaled condition number	any gas/encounter/spline	9.518027	any gas
events per parameter	any gas/encounter/spline	1448.688	any gas
nonevents per parameter	any gas/encounter/spline	1804.25	any gas
log likelihood	icd/encounter/spline	-30594.32	icd
null deviance	icd/encounter/spline	67657.92	icd
residual deviance	icd/encounter/spline	61188.64	icd
AIC	icd/encounter/spline	61220.64	icd
Brier	icd/encounter/spline	0.04395986	icd
log loss	icd/encounter/spline	0.1745544	icd
iterations	icd/encounter/spline	7	icd
residual df	icd/encounter/spline	175255	icd
scaled condition number	icd/encounter/spline	8.682502	icd
events per parameter	icd/encounter/spline	527.375	icd
nonevents per parameter	icd/encounter/spline	10427.06	icd

fit bmi min: 30; fit bmi max: 89.8000030517578; n observed outside: 0; maximum observed gap: 0; n unseen categories: 0; analysis population id: primary_bmi_ge30; population id: primary_bmi_ge30; ascertainment window: encounter; model form: spline.

Table 46: Curve and transformed prediction support — panel 1/2

model id	n reference	n transformed outside	pct transformed outside	maximum transformed gap	reference id	policy id	method
abg/encounter/spline	29021	2967	10.22363	0.8999996	C abg encounter	bmi gt30 minus1 v1	observed
any gas/encounter/spline	52047	5240	10.06782	0.8999996	C any gas encounter	bmi gt30 minus1 v1	observed
icd/encounter/spline	175271	14989	8.5519	0.8999996	C icd	bmi gt30 minus1 v1	observed
abg/encounter/spline	29021	0	0	0	C abg encounter	none; relationship	fixed BMI curve

model id	n reference	n transformed outside	pct transformed outside	maximum transformed gap	reference id	policy id	method
any gas/encounter/spline	52047	0	0	0	C any gas encounter	none; relationship	fixed BMI curve
icd/encounter/spline	175271	0	0	0	C icd	none; relationship	fixed BMI curve

Table 47: Curve and transformed prediction support — panel 2/2

model id	reference population id	outcome id
abg/encounter/spline	C abg encounter	abg
any gas/encounter/spline	C any gas encounter	any gas
icd/encounter/spline	C icd	icd
abg/encounter/spline	C abg encounter	abg
any gas/encounter/spline	C any gas encounter	any gas
icd/encounter/spline	C icd	icd

3.4 Other sensitivities and verification

Table 48: Fixed-index calendar-day transitions

outcome id	primary	day	n	analysis population id
abg	available negative	available negative	22012	primary
abg	positive	available negative	2577	primary
abg	unknown	available negative	0	primary
abg	available negative	positive	0	primary
abg	positive	positive	17196	primary
abg	unknown	positive	0	primary
abg	available negative	unknown	16316	primary
abg	positive	unknown	13217	primary
abg	unknown	unknown	308888	primary
any gas	available negative	available negative	54715	primary
any gas	positive	available negative	3861	primary
any gas	unknown	available negative	0	primary
any gas	available negative	positive	0	primary
any gas	positive	positive	33748	primary
any gas	unknown	positive	0	primary
any gas	available negative	unknown	19659	primary
any gas	positive	unknown	14091	primary
any gas	unknown	unknown	254132	primary

status: assessed; reason: same-row nested comparison; no model selection; df: 2; analysis population id: primary.

Table 49: Same-sample linear checks

outcome id	window	n model	likelihood ratio	p value	ascertainment window
abg	encounter	71318	231.7509	4.741738e-51	encounter
any gas	encounter	126074	410.6763	6.649395e-90	encounter
icd	encounter	380206	1438.521	4.256837e-313	encounter
abg	index calendar day	41785	103.2916	3.719823e-23	index calendar day
any gas	index calendar day	92324	287.6516	3.445553e-63	index calendar day

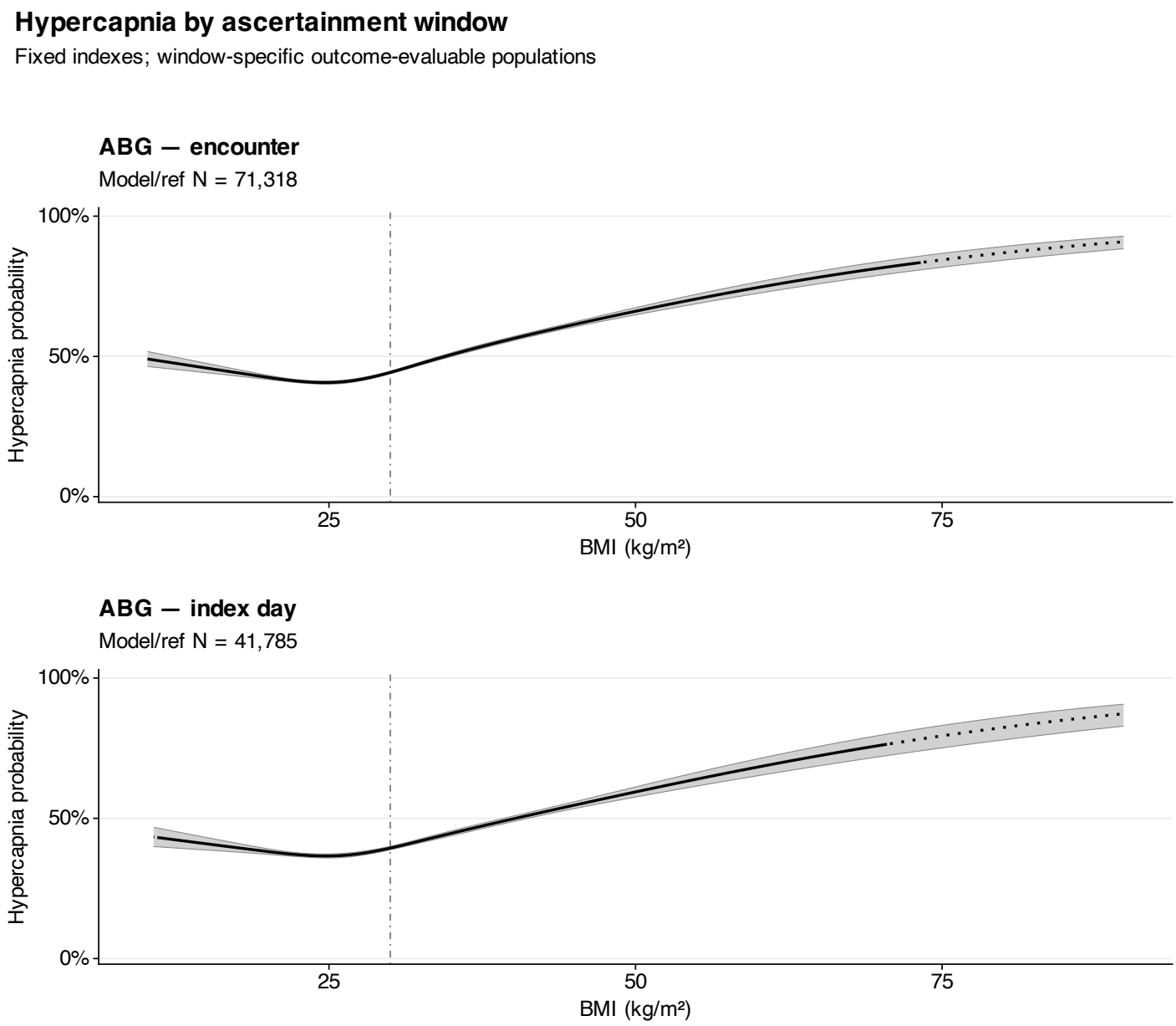


Figure 14: Analysis population: All valid BMI: primary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F5. Encounter versus encounter-start-calendar-day curves reuse the same indexes and fixed spline basis, on shared BMI/probability scales. Each curve standardizes over its displayed model/reference population. Bands are pointwise model-based 95% CIs; dotted segments are locally sparse and dashed segments exceed training BMI support. Pending source or failed models remain explicit. Index day is a calendar-date restriction, not elapsed 24 hours.

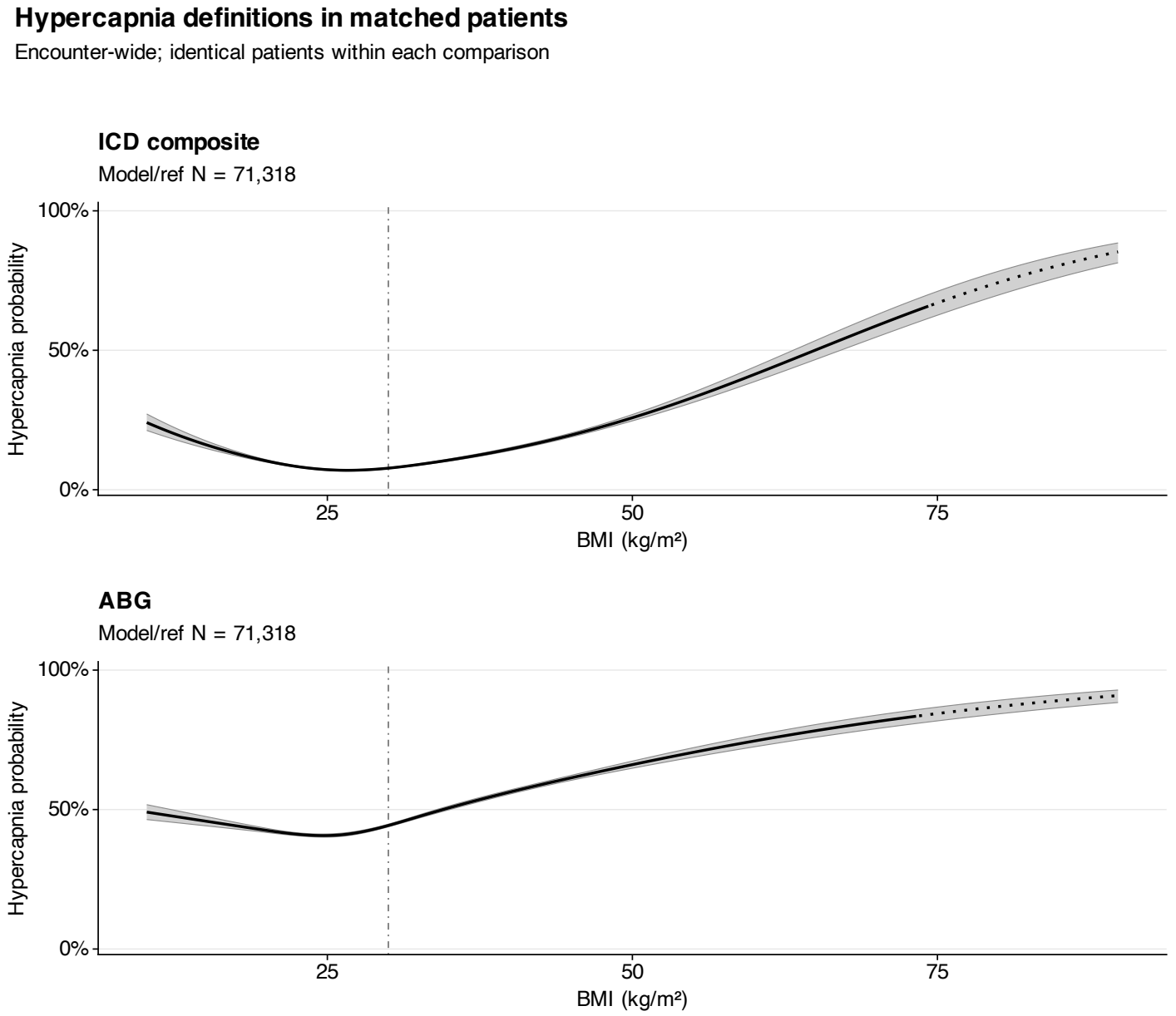


Figure 15: Analysis population: All valid BMI: primary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F5. Matched ICD/gas curves use identical training/reference patients within each pair and shared BMI/probability scales. Arterial and combined pairs can have different membership. Bands are pointwise model-based 95% CIs; dotted segments are locally sparse and dashed segments exceed training BMI support. ICD is the five-code composite including E66.2. Separate bands are not a test of between-model differences.

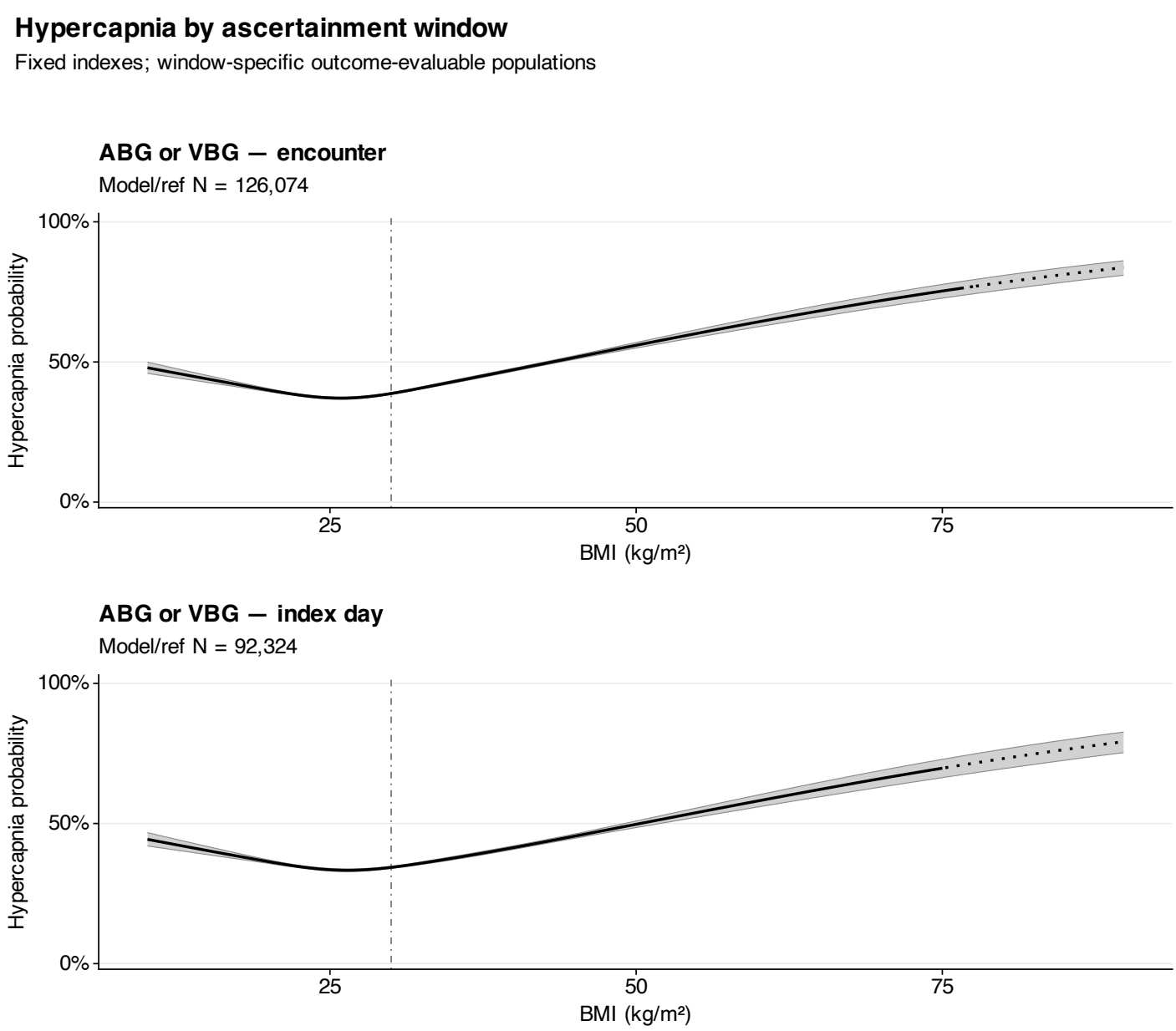


Figure 16: Analysis population: All valid BMI: primary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F5. Encounter versus encounter-start-calendar-day curves reuse the same indexes and fixed spline basis, on shared BMI/probability scales. Each curve standardizes over its displayed model/reference population. Bands are pointwise model-based 95% CIs; dotted segments are locally sparse and dashed segments exceed training BMI support. Pending source or failed models remain explicit. Index day is a calendar-date restriction, not elapsed 24 hours.

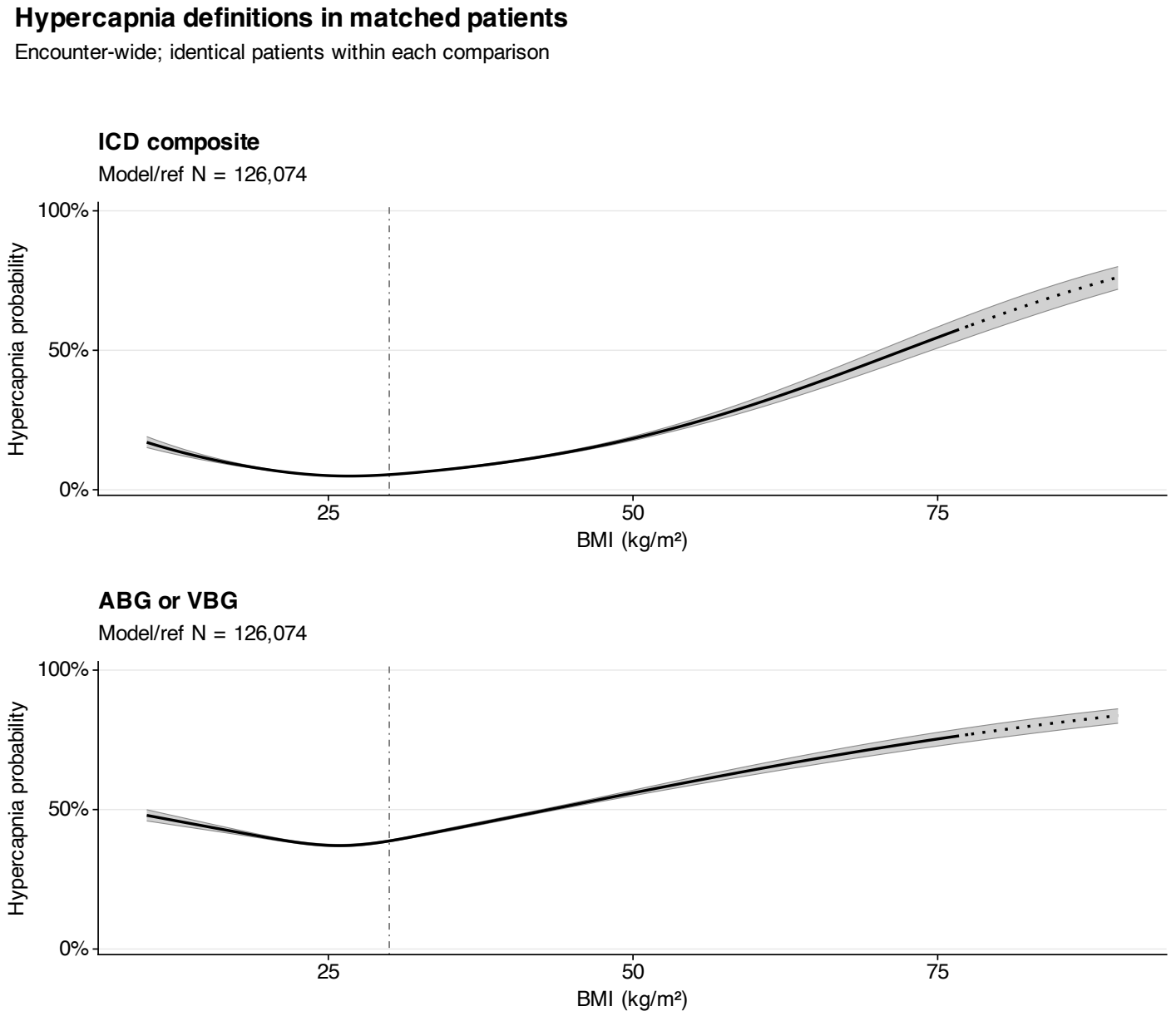


Figure 17: Analysis population: All valid BMI: primary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F5. Matched ICD/gas curves use identical training/reference patients within each pair and shared BMI/probability scales. Arterial and combined pairs can have different membership. Bands are pointwise model-based 95% CIs; dotted segments are locally sparse and dashed segments exceed training BMI support. ICD is the five-code composite including E66.2. Separate bands are not a test of between-model differences.

stage: Q.

Table 50: Primary BMI provenance and lags — panel 1/2

subgroup	method	bmi band	gas state	metric	n	known	unknown
non obesity only	calculated	40to50	negative	bmi lag days	2127	0	2127
non obesity only	calculated	40to50	negative	height lag days	2127	2127	0
non obesity only	calculated	40to50	negative	weight lag days	2127	2127	0
non obesity only	calculated	40to50	negative	component gap days	2127	2127	0
non obesity only	calculated	40to50	positive	bmi lag days	1953	0	1953
non obesity only	calculated	40to50	positive	height lag days	1953	1953	0
non obesity only	calculated	40to50	positive	weight lag days	1953	1953	0
non obesity only	calculated	40to50	positive	component gap days	1953	1953	0
non obesity only	calculated	40to50	unavailable	bmi lag days	6295	0	6295
non obesity only	calculated	40to50	unavailable	height lag days	6295	6295	0
non obesity only	calculated	40to50	unavailable	weight lag days	6295	6295	0
non obesity only	calculated	40to50	unavailable	component gap days	6295	6295	0
non obesity only	calculated	50plus	negative	bmi lag days	692	0	692
non obesity only	calculated	50plus	negative	height lag days	692	692	0
non obesity only	calculated	50plus	negative	weight lag days	692	692	0
non obesity only	calculated	50plus	negative	component gap days	692	692	0
non obesity only	calculated	50plus	positive	bmi lag days	1094	0	1094
non obesity only	calculated	50plus	positive	height lag days	1094	1094	0
non obesity only	calculated	50plus	positive	weight lag days	1094	1094	0
non obesity only	calculated	50plus	positive	component gap days	1094	1094	0
non obesity only	calculated	50plus	unavailable	bmi lag days	3571	0	3571
non obesity only	calculated	50plus	unavailable	height lag days	3571	3571	0
non obesity only	calculated	50plus	unavailable	weight lag days	3571	3571	0
non obesity only	calculated	50plus	unavailable	component gap days	3571	3571	0
non obesity only	calculated	below40	negative	bmi lag days	30874	0	30874
non obesity only	calculated	below40	negative	height lag days	30874	30874	0
non obesity only	calculated	below40	negative	weight lag days	30874	30874	0
non obesity only	calculated	below40	negative	component gap days	30874	30874	0
non obesity only	calculated	below40	positive	bmi lag days	20971	0	20971
non obesity only	calculated	below40	positive	height lag days	20971	20971	0
non obesity only	calculated	below40	positive	weight lag days	20971	20971	0
non obesity only	calculated	below40	positive	component gap days	20971	20971	0
non obesity only	calculated	below40	unavailable	bmi lag days	59826	0	59826

subgroup	method	bmi band	gas state	metric	n	known	unknown
non obesity only	calculated	below40	unavailable	height lag days	59826	59826	0
non obesity only	calculated	below40	unavailable	weight lag days	59826	59826	0
non obesity only	calculated	below40	unavailable	component gap days	59826	59826	0
non obesity only	current	40to50	negative	bmi lag days	778	778	0
non obesity only	current	40to50	negative	height lag days	778	759	19
non obesity only	current	40to50	negative	weight lag days	778	767	11
non obesity only	current	40to50	negative	component gap days	778	750	28
non obesity only	current	40to50	positive	bmi lag days	766	766	0
non obesity only	current	40to50	positive	height lag days	766	738	28
non obesity only	current	40to50	positive	weight lag days	766	758	8
non obesity only	current	40to50	positive	component gap days	766	731	35
non obesity only	current	40to50	unavailable	bmi lag days	6220	6220	0
non obesity only	current	40to50	unavailable	height lag days	6220	6127	93
non obesity only	current	40to50	unavailable	weight lag days	6220	6137	83
non obesity only	current	40to50	unavailable	component gap days	6220	6051	169
non obesity only	current	50plus	negative	bmi lag days	4	4	0
non obesity only	current	50plus	negative	height lag days	4	4	0
non obesity only	current	50plus	negative	weight lag days	4	4	0
non obesity only	current	50plus	negative	component gap days	4	4	0
non obesity only	current	50plus	positive	bmi lag days	3	3	0
non obesity only	current	50plus	positive	height lag days	3	3	0
non obesity only	current	50plus	positive	weight lag days	3	3	0
non obesity only	current	50plus	positive	component gap days	3	3	0
non obesity only	current	50plus	unavailable	bmi lag days	19	19	0
non obesity only	current	50plus	unavailable	height lag days	19	18	1
non obesity only	current	50plus	unavailable	weight lag days	19	19	0
non obesity only	current	50plus	unavailable	component gap days	19	18	1
non obesity only	current	below40	negative	bmi lag days	25836	25836	0
non obesity only	current	below40	negative	height lag days	25836	24543	1293
non obesity only	current	below40	negative	weight lag days	25836	24593	1243
non obesity only	current	below40	negative	component gap days	25836	23423	2413
non obesity only	current	below40	positive	bmi lag days	17861	17861	0
non obesity only	current	below40	positive	height lag days	17861	17067	794
non obesity only	current	below40	positive	weight lag days	17861	17261	600
non obesity only	current	below40	positive	component gap days	17861	16567	1294

subgroup	method	bmi band	gas state	metric	n	known	unknown
non obesity only	current	below40	unavailable	bmi lag days	105468	105468	0
non obesity only	current	below40	unavailable	height lag days	105468	101318	4150
non obesity only	current	below40	unavailable	weight lag days	105468	102682	2786
non obesity only	current	below40	unavailable	component gap days	105468	98831	6637
non obesity only	prior	40to50	negative	bmi lag days	876	876	0
non obesity only	prior	40to50	negative	height lag days	876	846	30
non obesity only	prior	40to50	negative	weight lag days	876	864	12
non obesity only	prior	40to50	negative	component gap days	876	839	37
non obesity only	prior	40to50	positive	bmi lag days	728	728	0
non obesity only	prior	40to50	positive	height lag days	728	713	15
non obesity only	prior	40to50	positive	weight lag days	728	721	7
non obesity only	prior	40to50	positive	component gap days	728	708	20
non obesity only	prior	40to50	unavailable	bmi lag days	8081	8081	0
non obesity only	prior	40to50	unavailable	height lag days	8081	7499	582
non obesity only	prior	40to50	unavailable	weight lag days	8081	7991	90
non obesity only	prior	40to50	unavailable	component gap days	8081	7443	638
non obesity only	prior	50plus	negative	bmi lag days	12	12	0
non obesity only	prior	50plus	negative	height lag days	12	12	0
non obesity only	prior	50plus	negative	weight lag days	12	12	0
non obesity only	prior	50plus	negative	component gap days	12	12	0
non obesity only	prior	50plus	positive	bmi lag days	15	15	0
non obesity only	prior	50plus	positive	height lag days	15	15	0
non obesity only	prior	50plus	positive	weight lag days	15	15	0
non obesity only	prior	50plus	positive	component gap days	15	15	0
non obesity only	prior	50plus	unavailable	bmi lag days	97	97	0
non obesity only	prior	50plus	unavailable	height lag days	97	96	1
non obesity only	prior	50plus	unavailable	weight lag days	97	92	5
non obesity only	prior	50plus	unavailable	component gap days	97	92	5
non obesity only	prior	below40	negative	bmi lag days	10162	10162	0
non obesity only	prior	below40	negative	height lag days	10162	9694	468
non obesity only	prior	below40	negative	weight lag days	10162	9873	289
non obesity only	prior	below40	negative	component gap days	10162	9580	582
non obesity only	prior	below40	positive	bmi lag days	4969	4969	0
non obesity only	prior	below40	positive	height lag days	4969	4770	199
non obesity only	prior	below40	positive	weight lag days	4969	4864	105

subgroup	method	bmi band	gas state	metric	n	known	unknown
non obesity only	prior	below40	positive	component gap days	4969	4727	242
non obesity only	prior	below40	unavailable	bmi lag days	40868	40868	0
non obesity only	prior	below40	unavailable	height lag days	40868	35856	5012
non obesity only	prior	below40	unavailable	weight lag days	40868	40049	819
non obesity only	prior	below40	unavailable	component gap days	40868	35382	5486
overlap	calculated	40to50	negative	bmi lag days	472	0	472
overlap	calculated	40to50	negative	height lag days	472	472	0
overlap	calculated	40to50	negative	weight lag days	472	472	0
overlap	calculated	40to50	negative	component gap days	472	472	0
overlap	calculated	40to50	positive	bmi lag days	572	0	572
overlap	calculated	40to50	positive	height lag days	572	572	0
overlap	calculated	40to50	positive	weight lag days	572	572	0
overlap	calculated	40to50	positive	component gap days	572	572	0
overlap	calculated	40to50	unavailable	bmi lag days	1956	0	1956
overlap	calculated	40to50	unavailable	height lag days	1956	1956	0
overlap	calculated	40to50	unavailable	weight lag days	1956	1956	0
overlap	calculated	40to50	unavailable	component gap days	1956	1956	0
overlap	calculated	50plus	negative	bmi lag days	212	0	212
overlap	calculated	50plus	negative	height lag days	212	212	0
overlap	calculated	50plus	negative	weight lag days	212	212	0
overlap	calculated	50plus	negative	component gap days	212	212	0
overlap	calculated	50plus	positive	bmi lag days	383	0	383
overlap	calculated	50plus	positive	height lag days	383	383	0
overlap	calculated	50plus	positive	weight lag days	383	383	0
overlap	calculated	50plus	positive	component gap days	383	383	0
overlap	calculated	50plus	unavailable	bmi lag days	1666	0	1666
overlap	calculated	50plus	unavailable	height lag days	1666	1666	0
overlap	calculated	50plus	unavailable	weight lag days	1666	1666	0
overlap	calculated	50plus	unavailable	component gap days	1666	1666	0
overlap	calculated	below40	negative	bmi lag days	401	0	401
overlap	calculated	below40	negative	height lag days	401	401	0
overlap	calculated	below40	negative	weight lag days	401	401	0
overlap	calculated	below40	negative	component gap days	401	401	0
overlap	calculated	below40	positive	bmi lag days	452	0	452
overlap	calculated	below40	positive	height lag days	452	452	0

subgroup	method	bmi band	gas state	metric	n	known	unknown
overlap	calculated	below40	positive	weight lag days	452	452	0
overlap	calculated	below40	positive	component gap days	452	452	0
overlap	calculated	below40	unavailable	bmi lag days	1349	0	1349
overlap	calculated	below40	unavailable	height lag days	1349	1349	0
overlap	calculated	below40	unavailable	weight lag days	1349	1349	0
overlap	calculated	below40	unavailable	component gap days	1349	1349	0
overlap	current	40to50	negative	bmi lag days	1279	1279	0
overlap	current	40to50	negative	height lag days	1279	1197	82
overlap	current	40to50	negative	weight lag days	1279	1230	49
overlap	current	40to50	negative	component gap days	1279	1150	129
overlap	current	40to50	positive	bmi lag days	1200	1200	0
overlap	current	40to50	positive	height lag days	1200	1145	55
overlap	current	40to50	positive	weight lag days	1200	1173	27
overlap	current	40to50	positive	component gap days	1200	1121	79
overlap	current	40to50	unavailable	bmi lag days	13545	13545	0
overlap	current	40to50	unavailable	height lag days	13545	12766	779
overlap	current	40to50	unavailable	weight lag days	13545	13230	315
overlap	current	40to50	unavailable	component gap days	13545	12483	1062
overlap	current	50plus	negative	bmi lag days	6	6	0
overlap	current	50plus	negative	height lag days	6	6	0
overlap	current	50plus	negative	weight lag days	6	5	1
overlap	current	50plus	negative	component gap days	6	5	1
overlap	current	50plus	positive	bmi lag days	2	2	0
overlap	current	50plus	positive	height lag days	2	2	0
overlap	current	50plus	positive	weight lag days	2	2	0
overlap	current	50plus	positive	component gap days	2	2	0
overlap	current	50plus	unavailable	bmi lag days	49	49	0
overlap	current	50plus	unavailable	height lag days	49	47	2
overlap	current	50plus	unavailable	weight lag days	49	47	2
overlap	current	50plus	unavailable	component gap days	49	45	4
overlap	current	below40	negative	bmi lag days	354	354	0
overlap	current	below40	negative	height lag days	354	346	8
overlap	current	below40	negative	weight lag days	354	329	25
overlap	current	below40	negative	component gap days	354	322	32
overlap	current	below40	positive	bmi lag days	450	450	0

subgroup	method	bmi band	gas state	metric	n	known	unknown
overlap	current	below40	positive	height lag days	450	442	8
overlap	current	below40	positive	weight lag days	450	418	32
overlap	current	below40	positive	component gap days	450	412	38
overlap	current	below40	unavailable	bmi lag days	2093	2093	0
overlap	current	below40	unavailable	height lag days	2093	2063	30
overlap	current	below40	unavailable	weight lag days	2093	2038	55
overlap	current	below40	unavailable	component gap days	2093	2010	83
overlap	prior	40to50	negative	bmi lag days	152	152	0
overlap	prior	40to50	negative	height lag days	152	149	3
overlap	prior	40to50	negative	weight lag days	152	152	0
overlap	prior	40to50	negative	component gap days	152	149	3
overlap	prior	40to50	positive	bmi lag days	166	166	0
overlap	prior	40to50	positive	height lag days	166	166	0
overlap	prior	40to50	positive	weight lag days	166	164	2
overlap	prior	40to50	positive	component gap days	166	164	2
overlap	prior	40to50	unavailable	bmi lag days	1723	1723	0
overlap	prior	40to50	unavailable	height lag days	1723	1624	99
overlap	prior	40to50	unavailable	weight lag days	1723	1718	5
overlap	prior	40to50	unavailable	component gap days	1723	1623	100
overlap	prior	50plus	negative	bmi lag days	3	3	0
overlap	prior	50plus	negative	height lag days	3	3	0
overlap	prior	50plus	negative	weight lag days	3	3	0
overlap	prior	50plus	negative	component gap days	3	3	0
overlap	prior	50plus	positive	bmi lag days	1	1	0
overlap	prior	50plus	positive	height lag days	1	1	0
overlap	prior	50plus	positive	weight lag days	1	1	0
overlap	prior	50plus	positive	component gap days	1	1	0
overlap	prior	50plus	unavailable	bmi lag days	28	28	0
overlap	prior	50plus	unavailable	height lag days	28	28	0
overlap	prior	50plus	unavailable	weight lag days	28	28	0
overlap	prior	50plus	unavailable	component gap days	28	28	0
overlap	prior	below40	negative	bmi lag days	134	134	0
overlap	prior	below40	negative	height lag days	134	133	1
overlap	prior	below40	negative	weight lag days	134	127	7
overlap	prior	below40	negative	component gap days	134	126	8

subgroup	method	bmi band	gas state	metric	n	known	unknown
overlap	prior	below40	positive	bmi lag days	114	114	0
overlap	prior	below40	positive	height lag days	114	113	1
overlap	prior	below40	positive	weight lag days	114	112	2
overlap	prior	below40	positive	component gap days	114	111	3
overlap	prior	below40	unavailable	bmi lag days	1278	1278	0
overlap	prior	below40	unavailable	height lag days	1278	1128	150
overlap	prior	below40	unavailable	weight lag days	1278	1246	32
overlap	prior	below40	unavailable	component gap days	1278	1109	169

Table 51: Primary BMI provenance and lags — panel 2/2

subgroup	median	maximum	status
non obesity only	unavailable	unavailable	unavailable or unknown
non obesity only	0	6679	assessed
non obesity only	0	6473	assessed
non obesity only	0	6574	assessed
non obesity only	unavailable	unavailable	unavailable or unknown
non obesity only	0	3457	assessed
non obesity only	0	5901	assessed
non obesity only	0	5901	assessed
non obesity only	unavailable	unavailable	unavailable or unknown
non obesity only	0	7545	assessed
non obesity only	0	7211	assessed
non obesity only	0	7325	assessed
non obesity only	unavailable	unavailable	unavailable or unknown
non obesity only	0	4164	assessed
non obesity only	0	6270	assessed
non obesity only	0	6255	assessed
non obesity only	unavailable	unavailable	unavailable or unknown
non obesity only	0	6641	assessed
non obesity only	0	6641	assessed
non obesity only	0	5160	assessed
non obesity only	unavailable	unavailable	unavailable or unknown
non obesity only	0	6817	assessed
non obesity only	0	7830	assessed
non obesity only	0	7830	assessed
non obesity only	unavailable	unavailable	unavailable or unknown
non obesity only	0	7064	assessed
non obesity only	0	6939	assessed
non obesity only	0	7064	assessed
non obesity only	unavailable	unavailable	unavailable or unknown
non obesity only	0	7608	assessed
non obesity only	0	7344	assessed
non obesity only	0	6775	assessed
non obesity only	unavailable	unavailable	unavailable or unknown
non obesity only	0	8039	assessed
non obesity only	0	7309	assessed
non obesity only	0	8039	assessed
non obesity only	0	0	assessed
non obesity only	0	2717	assessed
non obesity only	0	4165	assessed
non obesity only	0	4170	assessed
non obesity only	0	0	assessed
non obesity only	0	2646	assessed
non obesity only	0	3567	assessed
non obesity only	0	3569	assessed
non obesity only	0	0	assessed
non obesity only	0	6469	assessed
non obesity only	0	4192	assessed
non obesity only	0	6469	assessed

subgroup	median	maximum	status
non obesity only	-5	0	assessed
non obesity only	0	1749	assessed
non obesity only	0	0	assessed
non obesity only	4.5	1749	assessed
non obesity only	0	0	assessed
non obesity only	0	0	assessed
non obesity only	0	0	assessed
non obesity only	0	2	assessed
non obesity only	0	0	assessed
non obesity only	0	2	assessed
non obesity only	0	409	assessed
non obesity only	0	409	assessed
non obesity only	0	0	assessed
non obesity only	0	6944	assessed
non obesity only	0	4356	assessed
non obesity only	0	6944	assessed
non obesity only	0	0	assessed
non obesity only	0	6700	assessed
non obesity only	0	4372	assessed
non obesity only	0	6700	assessed
non obesity only	0	0	assessed
non obesity only	0	6984	assessed
non obesity only	0	4399	assessed
non obesity only	0	6984	assessed
non obesity only	190.5	6240	assessed
non obesity only	16	6240	assessed
non obesity only	3	5376	assessed
non obesity only	6	6119	assessed
non obesity only	186.5	6524	assessed
non obesity only	4	6641	assessed
non obesity only	1	4964	assessed
non obesity only	0	6482	assessed
non obesity only	141	6888	assessed
non obesity only	29	6756	assessed
non obesity only	16	6609	assessed
non obesity only	0	6756	assessed
non obesity only	201.5	2304	assessed
non obesity only	0	467	assessed
non obesity only	0	349	assessed
non obesity only	0	429	assessed
non obesity only	382	4581	assessed
non obesity only	1	1072	assessed
non obesity only	1	1072	assessed
non obesity only	0	326	assessed
non obesity only	357	3937	assessed
non obesity only	0	826	assessed
non obesity only	0	2534	assessed
non obesity only	0	2534	assessed
non obesity only	150	6635	assessed
non obesity only	21	6635	assessed
non obesity only	5	6669	assessed
non obesity only	19	5515	assessed
non obesity only	108	7231	assessed
non obesity only	11	5972	assessed
non obesity only	10	5934	assessed
non obesity only	1	5753	assessed
non obesity only	115	7729	assessed
non obesity only	69	7218	assessed
non obesity only	37	7729	assessed
non obesity only	0	6250	assessed
overlap	unavailable	unavailable	unavailable or unknown
overlap	0	4039	assessed
overlap	0	5859	assessed
overlap	0	5859	assessed
overlap	unavailable	unavailable	unavailable or unknown
overlap	0	4056	assessed
overlap	0	5712	assessed
overlap	0	5712	assessed
overlap	unavailable	unavailable	unavailable or unknown
overlap	0	5701	assessed
overlap	0	6495	assessed

subgroup	median	maximum	status
overlap	0	6441	assessed
overlap	unavailable	unavailable	unavailable or unknown
overlap	0	1761	assessed
overlap	0	5560	assessed
overlap	0	5560	assessed
overlap	unavailable	unavailable	unavailable or unknown
overlap	0	2286	assessed
overlap	0	5599	assessed
overlap	0	5599	assessed
overlap	unavailable	unavailable	unavailable or unknown
overlap	0	4505	assessed
overlap	0	5605	assessed
overlap	0	5605	assessed
overlap	unavailable	unavailable	unavailable or unknown
overlap	0	4404	assessed
overlap	0	6002	assessed
overlap	1	6002	assessed
overlap	unavailable	unavailable	unavailable or unknown
overlap	0	3386	assessed
overlap	0	5716	assessed
overlap	2	5716	assessed
overlap	unavailable	unavailable	unavailable or unknown
overlap	0	4851	assessed
overlap	0	6287	assessed
overlap	0	6293	assessed
overlap	0	0	assessed
overlap	0	6618	assessed
overlap	0	3739	assessed
overlap	0	6618	assessed
overlap	0	0	assessed
overlap	0	4569	assessed
overlap	0	3154	assessed
overlap	0	4569	assessed
overlap	0	0	assessed
overlap	0	6802	assessed
overlap	0	4434	assessed
overlap	0	6802	assessed
overlap	0	0	assessed
overlap	0	0	assessed
overlap	0	0	assessed
overlap	0	0	assessed
overlap	0	0	assessed
overlap	0	0	assessed
overlap	0	0	assessed
overlap	0	0	assessed
overlap	0	0	assessed
overlap	0	2322	assessed
overlap	0	0	assessed
overlap	0	2322	assessed
overlap	0	0	assessed
overlap	0	2798	assessed
overlap	0	3909	assessed
overlap	0	3909	assessed
overlap	0	0	assessed
overlap	0	2750	assessed
overlap	0	3844	assessed
overlap	0	3844	assessed
overlap	0	0	assessed
overlap	0	6976	assessed
overlap	0	3962	assessed
overlap	0	6976	assessed
overlap	533.5	4797	assessed
overlap	0	3649	assessed
overlap	168.5	3477	assessed
overlap	163	3289	assessed
overlap	583	5694	assessed
overlap	0	3227	assessed
overlap	0	3595	assessed
overlap	2	3595	assessed
overlap	489	7039	assessed
overlap	59	6769	assessed

subgroup	median	maximum	status
overlap	100	6455	assessed
overlap	0	4835	assessed
overlap	149	681	assessed
overlap	0	149	assessed
overlap	75	149	assessed
overlap	1	75	assessed
overlap	190	190	assessed
overlap	0	0	assessed
overlap	0	0	assessed
overlap	0	0	assessed
overlap	1060	4159	assessed
overlap	0	2421	assessed
overlap	1	1648	assessed
overlap	0	1659	assessed
overlap	344	4649	assessed
overlap	0	3485	assessed
overlap	160	3360	assessed
overlap	93	3360	assessed
overlap	339.5	3698	assessed
overlap	0	3698	assessed
overlap	136.5	3698	assessed
overlap	49	3552	assessed
overlap	156	6981	assessed
overlap	53	6981	assessed
overlap	69	6172	assessed
overlap	0	5433	assessed

3.4.1 Current BMI: sensitivity

family: supporting; window: encounter; model form: spline; source status: qualified; formula text: .outcome ~ bmi_ns1 + bmi_ns2 + bmi_ns3 + age_at_encounter + sex + race_ethnicity + location + encounter_type; basis id: 6094fd99aa4f5725f0a36afa82b83866f82625b4c83f87058d2bc53eafb813ec; adjustment set id: inherited_plus_encounter_setting_v1; weight specification: none; covariance method: model-based binomial; empirical reference fixed; n eligible: 278348; n excluded adjustment: 0; n parameters: 16; model rank: 16; status: successful; failure reason: ; warnings: ; analysis population id: primary_current; ascertainment window: encounter; population id: primary_current; source reason: unavailable; estimator: unavailable; separation detected: unavailable.

Table 52: Supporting fit status — panel 1/2

model id	outcome id	training population id	field	n excluded outcome	n model	n patients	n events
abg/encounter/spline	abg	C abg encounter	abg	222004	56344	56344	26198
any gas/encounter/spline	any gas	C any gas encounter	any gas	184653	93695	93695	39377
icd/encounter/spline	icd	C icd	icd	0	278348	278348	12087

Table 53: Supporting fit status — panel 2/2

model id	n nonevents	fit runtime
abg/encounter/spline	30146	0.307
any gas/encounter/spline	54318	0.514
icd/encounter/spline	266261	1.696

Maximum decrement: 1.

Table 54: Supporting scenario estimates — panel 1/2

Model	Reference	Model N	Reference N	Events	Patients shifted	Mean decrement	Eligible (%)
abg/encounter/spline	C abg encounter	56344	56344	26198	22218	0.3943277	39.43
any gas/encounter/ spline	C any gas encounter	93695	93695	39377	37332	0.3984418	39.84
icd/encounter/spline	C icd	278348	278348	12087	123541	0.4438365	44.38

Table 55: Supporting scenario estimates — panel 2/2

Model	Observed probability % (95% CI)	Shifted probability % (95% CI)	Difference per 1000 (95% CI)	PIF % (95% CI)
abg/encounter/spline	46.5 (46.09, 46.9)	46.02 (45.61, 46.43)	4.741 (4.44, 5.042)	1.02 (0.9541, 1.085)
any gas/encounter/ spline	42.03 (41.71, 42.34)	41.68 (41.36, 41.99)	3.505 (3.271, 3.738)	0.8339 (0.7781, 0.8897)
icd/encounter/spline	4.342 (4.268, 4.418)	4.228 (4.155, 4.301)	1.147 (1.083, 1.212)	2.642 (2.499, 2.784)

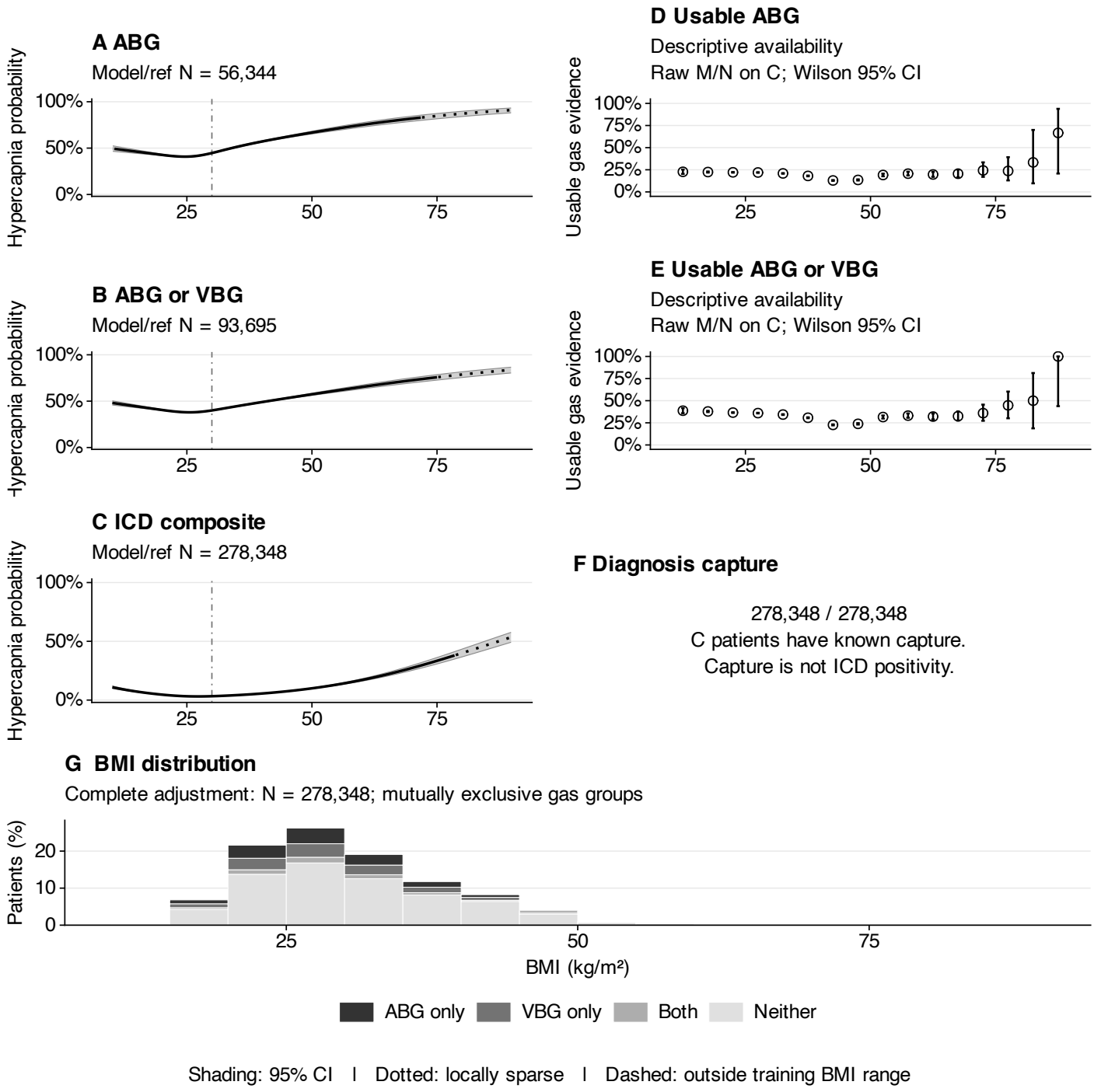


Figure 18: Analysis population: Current BMI: sensitivity. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F1. BMI–hypercapnia relationships at the first eligible adult 2022 acute-care encounter. ABG denotes $PaCO_2$ ≥45 mm Hg; ABG or VBG denotes $PaCO_2$ ≥45 or $PvCO_2$ ≥50 mm Hg. ICD denotes the five-code composite, including E66.2. The vertical dot-dash line marks BMI 30, the scenario eligibility boundary, not the cohort boundary. Gas curves standardize over outcome-evaluable patients; ICD uses known diagnosis capture. Model/ref N is the common model and reference count. Availability and BMI frequency use all adjustment-complete patients (C); the four frequency groups are mutually exclusive. Adjustment: age, sex, race/ethnicity, geographic region and encounter setting (EMER reference; IMP inpatient) with a fixed natural BMI spline. Shading and fine boundaries show pointwise model-based 95% CIs; dotted segments are locally sparse and dashed segments exceed training BMI support. Open availability points use Wilson intervals when no model is estimable. Known capture is distinct from ICD positivity. ABG/VBG: arterial/venous blood gas. Conditional model uncertainty excludes selection and measurement assumptions; no population transport or causal effect is inferred.

3.4.2 All OBESITY routes: overlapping sensitivity

family: supporting; window: encounter; model form: spline; source status: qualified; formula text: .outcome ~ bmi_ns1 + bmi_ns2 + bmi_ns3 + age_at_encounter + sex + race_ethnicity + location + encounter_type; basis id: b2402c75672e8d572228a099baaab46cc47d359d83e846b5bcf603eabce97f09; adjustment set id: inherited_plus_encounter_setting_v1; weight specification: none; covariance method: model-based binomial; empirical reference fixed; n eligible: 76913; n excluded adjustment: 0; n parameters: 16; model rank: 16; status: successful; failure reason: ; warnings: ; analysis population id: all_obesity; ascertainment window: encounter; population id: all_obesity; source reason: unavailable; estimator: unavailable; separation detected: unavailable.

Table 56: Supporting fit status — panel 1/2

model id	outcome id	training population id	field	n excluded outcome	n model	n patients	n events
abg/encounter/spline	abg	C abg encounter	abg	73045	3868	3868	2454
any gas/encounter/spline	any gas	C any gas encounter	any gas	70503	6410	6410	3365
icd/encounter/spline	icd	C icd	icd	0	76913	76913	2277

Table 57: Supporting fit status — panel 2/2

model id	n nonevents	fit runtime
abg/encounter/spline	1414	0.035
any gas/encounter/spline	3045	0.039
icd/encounter/spline	74636	0.47

Maximum decrement: 1.

Table 58: Supporting scenario estimates — panel 1/2

Model	Reference	Model N	Reference N	Events	Patients shifted	Mean decrement	Eligible (%)
abg/encounter/spline	C abg encounter	3868	3868	2454	3722	0.9622544	96.23
any gas/encounter/spline	C any gas encounter	6410	6410	3365	6195	0.9664587	96.65
icd/encounter/spline	C icd	76913	76913	2277	75794	0.9854511	98.55

Table 59: Supporting scenario estimates — panel 2/2

Model	Observed probability % (95% CI)	Shifted probability % (95% CI)	Difference per 1000 (95% CI)	PIF % (95% CI)
abg/encounter/spline	63.44 (61.93, 64.93)	62.98 (61.43, 64.49)	4.677 (2.308, 7.045)	0.7371 (0.3621, 1.111)
any gas/encounter/spline	52.5 (51.3, 53.69)	51.94 (50.73, 53.15)	5.538 (3.512, 7.563)	1.055 (0.6673, 1.441)
icd/encounter/spline	2.96 (2.847, 3.078)	2.811 (2.701, 2.925)	1.496 (1.29, 1.703)	5.055 (4.376, 5.729)

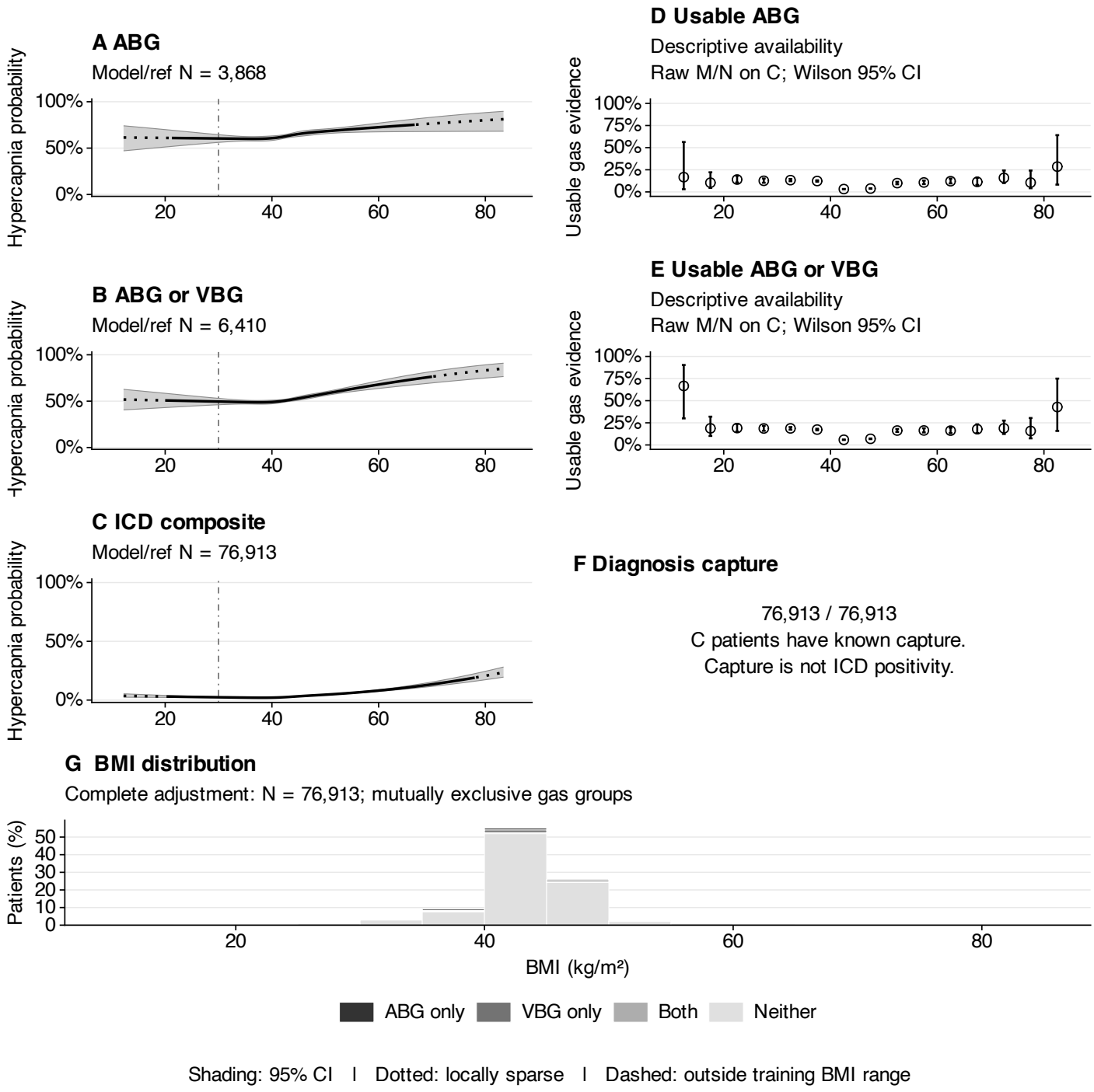


Figure 19: Analysis population: All OBESITY routes: overlapping sensitivity. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F1. BMI–hypercapnia relationships at the first eligible adult 2022 acute-care encounter. ABG denotes $PaCO_2$ ≥45 mm Hg; ABG or VBG denotes $PaCO_2$ ≥45 or $PvCO_2$ ≥50 mm Hg. ICD denotes the five-code composite, including E66.2. The vertical dot-dash line marks BMI 30, the scenario eligibility boundary, not the cohort boundary. Gas curves standardize over outcome-evaluable patients; ICD uses known diagnosis capture. Model/ref N is the common model and reference count. Availability and BMI frequency use all adjustment-complete patients (C); the four frequency groups are mutually exclusive. Adjustment: age, sex, race/ethnicity, geographic region and encounter setting (EMER reference; IMP inpatient) with a fixed natural BMI spline. Shading and fine boundaries show pointwise model-based 95% CIs; dotted segments are locally sparse and dashed segments exceed training BMI support. Open availability points use Wilson intervals when no model is estimable. Known capture is distinct from ICD positivity. ABG/VBG: arterial/venous blood gas. Conditional model uncertainty excludes selection and measurement assumptions; no population transport or causal effect is inferred.

The current-measurement restriction retains the same index and primary spline basis. All-OBESITY-route patients overlap the primary cohort; this analysis is not a partition or an independent comparison group. Obesity-only and unresolved source entry remain audit-only. No between-group test follows from separate intervals.

4 Interpretation

The primary relationship concerns all valid-BMI patients whose first qualifying encounter was represented through a non-OBESITY source pathway. The nested BMI ≥30 analysis addresses restriction at that same encounter. Broader BMI eligibility can change historical indexes and source-entry classifications; the new cohort is not merely the old cohort plus thinner patients.

The scenario changes exposure only above BMI 30 while retaining unshifted baseline burden below it. Gas findings remain conditional on evaluable evidence. Source entry, selective measurement, BMI timing, diagnostic incorporation, sparse support and residual confounding limit both observed relationships and model-based scenarios. Numerical reconciliation does not confer causal interpretation or investigator approval.

All applicable variables, coefficients, reference levels, covariance matrices, diagnostics, failures and supported contrasts—including 40 versus 45—are provided in the companion audit and aggregate tables. The plots describe relationships across BMI rather than privileging that interval.

Example interpretation for investigator review (not abstract text):

> **Example conclusion, pending investigator review:** We estimated BMI–hypercapnia probability relationships within the fixed-index, non-OBESITY-supported source-entry population. Outcome-specific contrasts and their uncertainty are reported separately; they do not constitute a between-group test. Gas-based estimates describe patients with usable blood-gas evidence in this selected acute-care cohort; diagnosis-based estimates describe patients with known diagnosis capture. The primary association remains subject to outcome-dependent entry and selective gas measurement. Secondary one-unit BMI-shift contrasts are descriptive model scenarios and do not establish effects of weight loss or cases prevented.